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Date 94・4・15

素粒子物理学

九後太一（汰一郎）通称　　レポート（5〜6、7回）
政池明

§0 SU(N)の群論

0-1. SU(2)（≃角運動量）　空間回転群 SO(3)

$g = \exp(i\theta^a J_a)$　　$[J_a, J_b] = i\epsilon_{abc} J_c$

↑ SU(2) Lie群　　generator　　Lie代数

群の表現：D次元表現

D成分ベクトル $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{pmatrix}$

$g: \vec{x} \to \vec{x}' = [\rho(g)]\vec{x}$　　such that　$\rho(g_1 g_2) = \rho(g_1)\rho(g_2)$

DxD行列

別の言い方

「V：D次元ベクトル空間　$g \in SU(2)$

$V \ni |x\rangle$ に対し（線形演算子を対応させる）表現

↳ $g|x\rangle$

base $\{|1\rangle, |2\rangle, \cdots, |D\rangle\}$

$|x\rangle = (|1\rangle, |2\rangle, \cdots, |D\rangle)\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{pmatrix}$

$g|i\rangle = \sum_j |j\rangle [\rho(g)]_{ji}$

$g|x\rangle = (|1\rangle, |2\rangle, \cdots, |D\rangle)\boxed{[\rho(g)]\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{pmatrix}}$

$\rho(g)$	$\rho(J_a)$	$\rho(g) = \exp(i\theta^a \rho(J_a))$
DxD行列	DxD行列	$[\rho(J_a), \rho(J_b)] = i\epsilon_{abc}\rho(J_c)$

基本表現（SO(3) スピノール表現）

2次元表現 $\{|1\rangle, |2\rangle\} = \{|\uparrow\rangle, |\downarrow\rangle\}$

$J^a|i\rangle = |j\rangle \left(\frac{\sigma_a}{2}\right)_{ji}$

$\Leftrightarrow |\psi\rangle = (|1\rangle, |2\rangle)\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}$

↳ スピノール spinor

σ_a：Pauli行列

$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$SU(2) \ni e^{i\theta_a J_a}$: $\psi \to \psi' = \underbrace{\left[e^{i\theta_a \frac{\sigma_a}{2}}\right]}_{\text{↳ SU(2) 全体}}\psi$

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$(e^X)^+ (e^X) = e^{X^+} e^X = e^{-X} e^X = 1$ $\quad SO(3) \simeq SU(2)/Z_2$

$X^+ = -X$

$\det(e^X) = e^{\mathrm{tr}X} = 1$ ⟩ special unitary 行列 $SU(2)$

既約表現

$$g(|1\rangle, |2\rangle, \cdots, |D_1\rangle, |D_1+1\rangle, \cdots |D\rangle) = (\overbrace{|1\rangle, \cdots |D_1\rangle}^{D_1}, \overbrace{|D_1+1\rangle, \cdots |D\rangle}^{D-D_1}) \begin{pmatrix} \overbrace{\rho_1(g)}^{D_1} & \overbrace{0}^{D-D_1} \\ 0 & \rho_2(g) \end{pmatrix}$$

ρ：可約 ⟷ 既約 $\qquad$ 完全可約 $\qquad$ コンパクト群

$SU(2) \simeq SO(3)$ の∀既約表現 = spin j 表現 $\quad j = \frac{1}{2}, 1, \frac{3}{2}, \cdots$

⇓

$(2j+1)$次元表現 $(|-j\rangle, |-j+1\rangle \cdots |+j\rangle)$

↓ j_zの固有値

全ての既約表現は基本表現 (spinor $j=\frac{1}{2}$) のテンソル積で作れる

$\{|i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_n\rangle = |i_1 i_2 \cdots i_n\rangle\}$ $(\frac{1}{2})$の空間のn重テンソル積空間 2^n次元

$|\psi\rangle = |i_1 i_2 \cdots i_n\rangle \psi_{i_1 i_2 \cdots i_n}$ 既約ではない

$g|i\rangle = |j\rangle \underbrace{[e^{i\theta_k \frac{\sigma_k}{2}}]_{ji}}_{U(g)}$

$g: \psi_{i_1 \cdots i_n} \to \psi'_{i_1 \cdots i_n} = [U(g)]_{i_1 j_1} [U(g)]_{i_2 j_2} \cdots [U(g)]_{i_n j_n} \psi_{j_1 \cdots j_n}$

例 $n=2$

$\psi_{ij} \to \psi'_{ij} = U_{ik} U_{jl} \psi_{kl}$ $\qquad \psi'_{ij} = U_{ik} U_{jl} \psi_{kl}$

$2^2 = 4$

対称. $\psi_{(ij)} = \psi_{ij} + \psi_{ji}$ ：3次元 $\qquad$ +) $\psi'_{ji} = U_{jk} U_{il} \psi_{kl} = U_{il} U_{jk} \psi_{kl} = U_{ik} U_{jl} \psi_{lk}$

反対称. $\psi_{[ij]} = \psi_{ij} - \psi_{ji}$ ：1次元 $\qquad \psi'_{(ij)} = U_{ik} U_{jl} \psi_{(kl)}$

spin 1 $\begin{cases} +1 & \psi_{(11)} \\ 0 & \psi_{(12)} = \psi_{(21)} \\ -1 & \psi_{(22)} \end{cases}$ $\qquad$ spin 0 : $\psi_{[12]}$ $\qquad$ $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$

$\psi_{(ij)} = \mathcal{S}(12) \psi_{ij} = (e + (12)) \psi_{ij}$ $\qquad (12) = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$

$\psi_{[ij]} = \mathcal{A}(12) \psi_{ij} = (e - (12)) \psi_{ij}$

置換群：(対称群) $S_n \ni \sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & & \sigma(n) \end{pmatrix}$

$\sigma \psi_{i_1 i_2 \cdots i_n} = \psi_{i_{\sigma(1)} i_{\sigma(2)} \cdots i_{\sigma(n)}}$

Young Symmetrizer

$\mathcal{Y}_{\boxed{\begin{smallmatrix}1&3\\2&4\end{smallmatrix}}} = [\mathcal{A}(14)\mathcal{A}(25)][\mathcal{S}(45)\mathcal{S}(123)]$ $\to \underbrace{e + \cdots + \cdots + \cdots}_{3!}$

$\mathcal{Y}_{\square\square} \psi_{ij}$

$\mathcal{Y}_{\square} \psi_{ij}$

$\left(\mathcal{Y}_{\square\square\square\square} \psi_{i_1 \cdots i_n}\right)$ 既約な成分

0-2 SU(N)

(1) 基本表現 $|\psi\rangle = \sum_{i=1}^{N} |i\rangle \psi_i = (|1\rangle, |2\rangle, \cdots, |N\rangle) \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_N \end{pmatrix}$

$SU(N) \ni g \; ; \; \psi_i \to \psi_i' = \underbrace{[\rho_N(g)]_i{}^j}_{\| \; U(g)} \psi_j$

$N \times N$ ユニタリー行列 (unitary ↓ $SU(N)$ ↑ special) $\det = 1$

(2) $\forall$既約表現 $\leftrightarrow$ 基本表現のテンソル積を Young symmetrizer で"対称化"したもの

$Y_{\square}$

例 $\psi_{\boxed{\begin{smallmatrix}i&j\\k\end{smallmatrix}}} = Y_{\square\square} \psi_{ijk}$ (N^3)

$\psi_{(ijk)} = \psi_{\boxed{i}\boxed{j}\boxed{k}} = Y_{\square\square\square} \psi_{ijk}$

$\psi_{[ijk]} = \psi_{\begin{smallmatrix}\boxed{i}\\\boxed{j}\\\boxed{k}\end{smallmatrix}} = Y_{\begin{smallmatrix}\square\\\square\\\square\end{smallmatrix}} \psi_{ijk}$

ψ : SU(N)不変 1次元表現 ／ $\square : \psi_i$ (N) ／ $\begin{smallmatrix}\square\\\square\end{smallmatrix} : \psi_{[ij]}$ (${}_NC_2$) ／ $\begin{smallmatrix}\square\\\square\\\square\end{smallmatrix} : \psi_{[ijk]}$ ／ $\cdots$ ／ $N-1$ 段 : $\psi_{[i_1 i_2 \cdots i_{N-1}]}$ (${}_NC_{N-1} = N^*$) ／ N 段 : $\psi_{[i_1 i_2 \cdots i_N]}$ 1次元 SU(N)不変

(3) 複素共役表現

∃表現 R : $\phi' = [\rho_R(g)]\phi$

$\updownarrow$

複素共役表現 R^* ($\bar{R}$) : $\bar{\phi}' = [\rho_{R^*}(g)]\bar{\phi} = (\rho_R(g))^*$ $\quad \bar{\phi} \sim \phi^*$

$(\psi_j)^* = \underline{\underline{\psi^i}}$ (N^*表現)

$*$ $\psi_i' = U_i{}^j \psi_j$

$\psi'^i = (U_i{}^j)^* \psi^j$

$\psi^i \sim \epsilon^{i j_1 \cdots j_{N-1}} \psi_{[j_1 \cdots j_{N-1}]}$

↓ N階完全反対称テンソル $\epsilon^{12\cdots N} = 1$

U

$\epsilon^{i j_1 \cdots j_{N-1}} U_{j_1}{}^{k_1} U_{j_2}{}^{k_2} \cdots U_{j_{N-1}}{}^{k_{N-1}} \psi_{[k_1 \cdots k_{N-1}]}$

$= (U_i{}^j)^* \epsilon^{j k_1 \cdots k_{N-1}} \psi_{[k_1 \cdots k_{N-1}]}$

($\because$) $\epsilon^{i_1 \cdots i_N} U_{i_1}{}^{j_1} U_{i_2}{}^{j_2} \cdots U_{i_N}{}^{j_N} = \det U \, \epsilon^{j_1 \cdots j_N}$

($\epsilon^{i_1 \cdots i_N}$: sgn ／ $U_{i_1}{}^{j_1}$: $\sigma(1)$ ／ $U_{i_2}{}^{j_2}$: $\sigma(2)$)

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N 既約表現 ⟷ 田

N-1 { □□□□□ / □□□□ / □□□ / □□ }

N { □ m / □ N-m } ⟲ *

(4) 既約表現の次元

一般の場合 factor over hook rule

次元 = F/N

F = N | N+1 | N+2 → +1 / N-1 | N / N-2 ↓ -1

H = 5 | 3 | 1 / 3 | 1 / 1

n

N | N+1 | ····· | N+n-1 / n | n-1 | ····· | 1 = N+n-1 C n = N H n

n

1 ··· 1 (1/2 1/2) SU(2) n+1 C n = n+1

spin j = n/2

-j, ···, j : 2j+1

SU(3) : □ : 3

εijk 田 ~ □* : 3

田 : 8 実表現 

(5) 既約表現の積の既約分解 ~ (SU(2) のスピン合成)

3 × 3* = 8 + 1

ψ□ · φ□

ψ(ij) φk

□□ × □ = □□□ + 

SU(2) (j=1)¹ × (1/2)² = 3/2 + 1/2

SU(3) 6 × 3 = 10 + 8

田 × □ =  + 

3* × 3 = 8 + 1

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§1 Quark Model

数百 hadrons { Meson (ボーズ粒子) → スピン整数
Baryon (フェルミ粒子) → スピン半整数

1.1 lowest lying hadrons

1) Meson

$J^{PC} = 0^{-+}$ pseudo-scalar メソン

$I=1$ アイソスピン: $\begin{pmatrix}\pi^+\\ \pi^0\\ \pi^-\end{pmatrix}$ ~ 140 MeV

$I=\frac{1}{2}$: $\begin{pmatrix}K^+\\ K^0\end{pmatrix}\begin{pmatrix}\bar{K}_0\\ K^-\end{pmatrix}$ ~ 494, ~ 497

$I=0$: η ~ 548, η' ~ 958 → 1

$\underset{\sim}{8}$ + 1

$J^{PC} = 1^{--}$ vector meson

$I=1$: $\begin{pmatrix}\rho^+\\ \rho^0\\ \rho^-\end{pmatrix}$ ~ 770

$I=\frac{1}{2}$: $\begin{pmatrix}K^{*+}\\ K^{*0}\end{pmatrix}\begin{pmatrix}\bar{K}^{*0}\\ K^{*-}\end{pmatrix}$ ~ 892, ~ 899

$I=0$: ω ~ 783, ϕ ~ 1020 → 1

$\underset{\sim}{8}$ + 1

$e \sim 0.5$ MeV

2) Baryon

$J^P = \frac{1}{2}^+$ octet baryon

$I=\frac{1}{2}$: $\begin{pmatrix}p\\ n\end{pmatrix}$ ~ 938, 939

$I=0$: Λ ~ 1116

$I=1$: $\begin{pmatrix}\Sigma^+\\ \Sigma^0\\ \Sigma^-\end{pmatrix}$ ~ 1189, 1192, 1197

$I=\frac{1}{2}$: $\begin{pmatrix}\Xi^0\\ \Xi^-\end{pmatrix}$ ~ 1314, 1321

$\underset{\sim}{8}$

$J^P = \frac{3}{2}^+$ decuplet

$I=\frac{3}{2}$: $\begin{pmatrix}\Delta^{++}\\ \Delta^+\\ \Delta^0\\ \Delta^-\end{pmatrix}$ ~ 1230

+150

$I=1$: $\begin{pmatrix}\Sigma^{*+}\\ \Sigma^{*0}\\ \Sigma^{*-}\end{pmatrix}$ ~ 1380

+150

$I=\frac{1}{2}$: $\begin{pmatrix}\Xi^{*0}\\ \Xi^{*-}\end{pmatrix}$ ~ 1532, 1535

+150? (Ω^- の予言)

$I=0$: Ω^- ~ 1672

$\underset{\sim}{10}$

▷ May 6

1.2 quark model : $SU(3)_{flavor}$

交換子 $[A, B] = AB - BA$

アンチ〃 $\{A, B\} = AB + BA$

$\begin{pmatrix}u\\ d\\ s\end{pmatrix} = \begin{pmatrix}q_1\\ q_2\\ q_3\end{pmatrix} = q_i \quad i = 1, 2, 3$

{ creation op. $q^{\dagger i}$ $(\bar{q}^{\dagger}_i)$ anti-quark
annihilation op. q_i $(\bar{q}^i)$ $\{q_i, q^{\dagger j}\} = \delta_i^{\ j}$

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$\frac{F}{H} = \frac{\square\square}{\square} = 8$

Meson $|M^i{}_j\rangle = q^{\dagger i}\,\bar{q}^{\dagger}{}_j\,|0\rangle$

$\square \times \overline{\square} = \square\!\square + \square$ (3 × 3* = 8 + 1) Ps, V 2重

Baryon $|B^{ijk}\rangle = q^{\dagger i}\,q^{\dagger j}\,q^{\dagger k}\,|0\rangle$

$\square \times \square \times \square = (\square\!\square + \square) \times \square$

$= \square\!\square\!\square + \square\!\square + \square\!\square + \square$ (10 + 8 + 8 + 1)

何故ここだけ？

→ SU(6) が答えを与える

quark の量子数

	I	I_z	Q (電荷)	S (Strangeness)	B (Baryon数)
u	1/2	+1/2	+2/3	0	+1/3
d		−1/2	−1/3	0	+1/3
s	0	0	−1/3	−1	+1/3

例 Decuplet $\square\!\square\!\square = \boxed{s|s|s} + \boxed{s|s|l} + \boxed{s|l|l} + \boxed{l|l|l}$ ← SU(2) isospin

$\psi_{(ijk)}$

1	2	3	4
I = 0	$I = \frac{1}{2}$	I = 1	$I = \frac{3}{2}$
$\Omega^- \,\|(sss)\rangle$	$\Xi^{*0}\,\|(ssu)\rangle$	$\Sigma^{*+}\,\|(suu)\rangle$	$\|(uuu)\rangle = \Delta^{++}$
	$\Xi^{*-}\,\|(ssd)\rangle$	$\Sigma^{*0}\,\|(sud)\rangle$	$\|(uud)\rangle = \Delta^{+}$
		$\Sigma^{*-}\,\|(sdd)\rangle$	$\|(udd)\rangle = \Delta^{0}$
			$\|(ddd)\rangle = \Delta^{-}$

問1 Baryon Octet $\square\!\square = \boxed{s|s} + \boxed{\begin{smallmatrix}s\\|\\l\end{smallmatrix}} + \boxed{\begin{smallmatrix}s\\|\\l\end{smallmatrix}\!\!-\!\!l} + \square\!\square$ (s方向反対称)

$I = \frac{1}{2}$ $I = 0$ $I = 1$ $I = \frac{1}{2}$

これらの状態を考えて 量子数 Q, I, I_z を調べて

p, n, $\Sigma^{\pm,0}$, $\Xi^{0,-}$, Λ との対応を付け

$\left(\boxed{\begin{smallmatrix}j\,i\\|\\k\end{smallmatrix}} \Longleftrightarrow \epsilon_{ijk}\,q^{\dagger i}\,q^{\dagger j}\,q^{\dagger k}\,|0\rangle - \frac{1}{3}\,\delta_i{}^{i}\,\epsilon_{mjk}\,q^{\dagger m}\,q^{\dagger j}\,q^{\dagger k}\,|0\rangle = |B_8{}^{i}{}_{j}\rangle \right)$

問2. $|M^i{}_j\rangle = q^{\dagger i}\,\bar{q}^{\dagger}{}_j\,|0\rangle \to 8 + 1$

↑ $q^{\dagger i}\,\bar{q}^{\dagger}{}_i\,|0\rangle$

の状態の Q, I, I_z を調べて Pseudo scalar meson との対応を考えよ

1.3 SU(6)

$\begin{pmatrix} u\uparrow \\ u\downarrow \\ d\uparrow \\ d\downarrow \\ s\uparrow \\ s\downarrow \end{pmatrix} = q_{i\alpha} = q_I$ (flavor, spin) $i=1,2,3$ $\alpha=1,2$

Meson $|M^I{}_J\rangle = q^{\dagger I}\,\bar{q}^{\dagger}{}_J\,|0\rangle$

Baryon $|B^{IJK}\rangle = q^{\dagger I}\,q^{\dagger J}\,q^{\dagger K}\,|0\rangle$

Meson $\square \times \square^* = \square\!\square + \square$

$\underset{\sim}{6} \times \underset{\sim}{6^*} = \underset{\sim}{35} + \underset{\sim}{1}$

$SU(6) \longleftrightarrow (SU(3)_{\text{flavor}},\ SU(2)_{\text{spin}})$

$\square \longleftrightarrow (\square, \square) \overset{*}{\longleftrightarrow} (\overline{\square\!\square}^{3^*}, \square^{2})$

$(\square,\square)\times(\overline{\square\!\square},\square) = (\square\!\square + \overline{\square\!\square},\ \square\!\square + \overline{\square\!\square})$

$= \overbrace{(\square\!\square,\square\!\square) + (\square\!\square,\overline{\square\!\square}) + (\overline{\square\!\square},\square\!\square)}^{35} + \overbrace{(\overline{\square\!\square},\overline{\square\!\square})}^{1}$

$\underset{\sim}{8}\ J=1 \qquad \underset{\sim}{8}\ J=0 \qquad \underset{\sim}{1}\ J=1 \qquad \underset{\sim}{1}\ J=0$

Vector 8+1 Pseudo-scalar

完全対称状態のみ ↓ 実現している！

Baryon $\square \times \square \times \square = \boxed{\square\!\square\!\square} + \square\!\square + \square\!\square + \overline{\square\!\square}$

$\underset{\sim}{6}\ \underset{\sim}{6}\ \underset{\sim}{6} \qquad \underset{\sim}{56} \qquad \underset{\sim}{70} \qquad \underset{\sim}{70} \qquad \underset{\sim}{20}$

spin $\frac{1}{2}$ のスピン 反対称は できない..

$(\square,\square)\times(\square,\square)\times(\square,\square) = (\square\!\square\!\square + \square\!\square + \square\!\square + \overline{\square\!\square},\ \square\!\square\!\square + \square\!\square + \square\!\square + \cancel{\overline{\square\!\square}})$

$SU(6)\ \underset{\sim}{56} = (\square\!\square\!\square, \square\!\square\!\square) + (\square\!\square, \square\!\square)$

$\underset{\sim}{10}\ J=\frac{3}{2} \qquad \underset{\sim}{8}\ J=\frac{1}{2}$

$q^\dagger$ は ボソン？ (実はウソ)

$J=\frac{1}{2}$ Dirac場 ⟷ fermion

boson

$[a_i, a^{\dagger j}] = \delta_i^j$ $\qquad a^{\dagger 1}a^{\dagger 2}|0\rangle = |1\,2\rangle + |2\,1\rangle$

$[a_i, a_j] = 0$ $\qquad a^{\dagger 2}a^{\dagger 1}|0\rangle$

$[a^{\dagger i}, a^{\dagger j}] = 0$

$[\,,\,] \to \{\,,\,\}$ $\qquad a^{\dagger 1}a^{\dagger 2}|0\rangle = |1\,2\rangle - |2\,1\rangle$

fermion $\qquad a^{\dagger 2}a^{\dagger 1}|0\rangle = -|2\,1\rangle$

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$q_{Ia} = q_{(i\alpha)a}$

flavor, spin, color　$a = 1,2,3$　SU(3) color

$|B^{IJK}\rangle = \epsilon_{abc}\, q^{\dagger Ia}\, q^{\dagger Jb}\, q^{\dagger Kc}\,|0\rangle$　→ color の自由度 の存在」

fermi 統計
$\{q_{Ia}, q^{\dagger Jb}\} = \delta_I^J \delta_a^b$
$\{q^{\dagger Ia}, q^{\dagger Jb}\} = 0$
$\{q_{Ia}, q_{Jb}\} = 0$

→ 強い相互作用」= color について singlet 状態のみが実現する

$|M^I{}_J\rangle = q^{\dagger Ia}\, \bar{q}^{\dagger}{}_{Ja}\,|0\rangle$

color singlet

$|M^I{}_J\rangle = q^{\dagger Ia}\, \bar{q}^{\dagger}{}_{Ja}\,|0\rangle$

color の閉じ込め

▷ May 13

2. 自由度の量子化　場の理論

2-1 実スカラー場 $\phi(x)$

$\mathcal{L}(\phi, \partial_\mu\phi) = \frac{1}{2}(\underbrace{\partial_\mu\phi\,\partial^\mu\phi}_{\text{運動項}} - \underbrace{m^2\phi^2}_{\text{質量項}}) - \underbrace{\frac{\lambda}{4!}\phi^4}_{\text{相互作用項}}$

$\partial_\mu = \frac{\partial}{\partial x^\mu} = (\frac{\partial}{\partial t}, \nabla)$　$\partial^\mu = (\frac{\partial}{\partial t}, -\nabla)$

$x^\mu = (t, \vec{x}) = (x^0, x^1, x^2, x^3)$　$c=1$, $\hbar=1$

$g^{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix} = g_{\mu\nu}$

$x_\mu = g_{\mu\nu} x^\nu = (t, -\vec{x})$

$x \cdot y = xy = x^\mu y_\mu = x_\mu y^\mu = x_0 y_0 - \vec{x}\cdot\vec{y}$

$\frac{\partial}{\partial x^\mu}(xy) = y_\mu$

$\frac{\partial}{\partial x_\mu}(xy) = y^\mu$

$S = \int d^4x\, \mathcal{L}(\phi, \partial_\mu\phi)$

$L = \int d^3x\, \mathcal{L}(\phi(x,t), \partial_\mu\phi(x,t))$

$= \int d^3x\, \frac{1}{2}\left[\dot{\phi}^2(x,t) - (\nabla\phi(x,t))^2 - m^2\phi^2(x,t) - \frac{\lambda}{4!}\phi^4(x,t)\right]$　$\partial_0\phi = \frac{\partial\phi}{\partial t} = \dot{\phi}$

t: fix
$= \int d^3x\, \frac{1}{2}\left[\dot{\phi}^2(x,t) - V(\phi(x,t))\right]$　$V(\phi) = \frac{1}{2}\left[(\nabla\phi)^2 + m^2\phi^2 + \frac{\lambda}{4!}\phi^4\right]$

$L = \sum_i \left[\frac{1}{2}\dot{q}_i^2 - V(q_i)\right]$　$p_i = \frac{\partial L}{\partial \dot{q}_i} = \dot{q}_i$　$i \to x$

$H = \sum_i p_i \dot{q}_i - L = \sum_i \left[\frac{1}{2}p_i^2 + V(q_i)\right]$　$q_i \to \phi(x)$

量子化
$[q_i, p_j] = i\hbar\,\delta_{ij}$
$[q_i, q_j] = [p_i, p_j] = 0$

$\frac{\partial q_i}{\partial q_j} = \delta_{ij} \to \frac{\delta\phi(y)}{\delta\phi(x)} = \delta^3(y-x)$

運動量変数 $\pi(y,t) = \dfrac{\delta L}{\delta \dot\phi(y,t)} = \dot\phi(y,t)$

$H = \int d^3x \left[\pi(x,t)\dot\phi(x,t) \right] - L$

量子化
$$\begin{cases} [\phi(x,t), \pi(y,t)] = i\,\delta^3(x-y) \\ [\phi(x,t), \phi(y,t)] = 0 \\ [\pi(x,t), \pi(y,t)] = 0 \end{cases} \qquad \pi(x) = -i\frac{\delta}{\delta\phi(x)}$$

$V(\phi) = \frac{1}{2}\left((\nabla\phi)^2 + m^2\phi^2\right) + \frac{\lambda}{4!}\phi^4$

$\downarrow \qquad\qquad \downarrow$

$-\nabla^2\phi \quad + m^2\phi + \frac{\lambda}{3!}\phi^3$

Heisenberg 運動方程式

$i\dot\phi(x,t) = [\phi(x,t), H] = i\pi(x,t)$

$i\dot\pi(x,t) = [\pi(x,t), H] = i\left[(\nabla^2 - m^2)\phi^2 - \frac{\lambda}{3!}\phi^3\right]$

$\dot\pi = \partial_0^2\phi$

$\rightarrow \quad (\partial_0^2 - \nabla^2 + m^2)\phi = -\frac{\lambda}{3!}\phi^3$

$(\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3$

$\partial^\mu\partial_\mu = \partial_0^2 - \nabla^2 = \Box$ d'Alembertian

$\mathcal{L} = \frac{1}{2}\left((\partial_\mu\phi)^2 - m^2\phi^2\right) - \frac{\lambda}{4!}\phi^4$

$-\partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} + \frac{\partial\mathcal{L}}{\partial\phi} = -\Box\phi - m^2\phi - \frac{\lambda}{3!}\phi^3 = 0$

自由場の場合 ($\lambda = 0$) λ: 結合定数

$(\Box + m^2)\phi = 0$: Klein-Gordon 方程式

$\phi(x,t) = \int \frac{d^3k}{(2\pi)^{3/2}} e^{ikx}\, g_k(t)$

$\partial_0^2 - \nabla^2 + m^2 \qquad \ddot g_k + \underbrace{(k^2+m^2)}_{k_0^2} g_k = 0 \qquad k_0 = \sqrt{k^2+m^2} > 0$

一般解 $g_k(t) = g_1(k)\, e^{-ik_0t} + g_2(k)\, e^{ik_0t}$ $\qquad k\cdot x = k_0t - k\cdot x$

$\phi(x,t) = \int \frac{d^3k}{(2\pi)^{3/2}}\left[g_1(k)\, e^{-ik\cdot x} + \underbrace{g_2(-k)}_{g_1^+(k)} e^{ikx}\right]$ $\qquad g_1(k) = \frac{1}{\sqrt{2k_0}}a(k)$ と書く

↓ real

$\phi(x) = \int \frac{d^3k}{\sqrt{(2\pi)^3 2k_0}}\left[a(k)\, e^{-ikx} + a^+(k)\, e^{ikx}\right]$

問1. これを導け

$a(k) = i\int d^3x\, f_k^*(x) \overleftrightarrow{\partial_0} \phi(x)$

$a^+(k) = -i\int d^3x \underbrace{f_k(x) \overleftrightarrow{\partial_0} \phi(x)}_{f_k\dot\pi - \dot f_k\phi}$

$f \overleftrightarrow{\partial_\mu} g = f\partial_\mu g - (\partial_\mu f)g$

$f_k(x) = \frac{1}{\sqrt{(2\pi)^3 2k_0}} e^{-ikx}$

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→ $[a(k), a^\dagger(q)] = \delta^3(k-q)$

$[a(k), a(q)] = [a^\dagger(k), a^\dagger(q)] = 0$

エネルギー・運動量　　$H = \int d^3x\,(\pi\dot\phi - \mathcal{L})$　　$P = \int d^3x\,(\pi\nabla\phi)$

$P^\mu = \int d^3x\,(\pi\partial^\mu\phi - g^{\mu 0}\mathcal{L})$ ← Noether の定理

$\int d^3x\, e^{ik\cdot x} = (2\pi)^3\delta^3(k)$

$= \int d^3k\,\frac{k^\mu}{2}\left[a^\dagger(k)a(k) + a(k)a^\dagger(k)\right]$　　$k^\mu = (\sqrt{k^2+m^2}, k)$

$a^\dagger(k)a(k) + \delta^3(0)\int\frac{d^3x}{(2\pi)^3}$

問2. この計算をせよ

$= \left(\int d^3k\, k^\mu\, a^\dagger(k)a(k)\right) + g^{\mu 0}\int\frac{d^3k\,d^3x}{(2\pi\hbar)^3}\cdot\frac{\hbar}{2}\underbrace{\sqrt{k^2+m^2}}_{\omega}$

$[P^\mu, a^\dagger(k)] = k^\mu a^\dagger(k)$　→　$P^\mu a^\dagger = a^\dagger(P^\mu + k^\mu)$

真空 $|0\rangle$: $a(k)|0\rangle = 0$

$P^\mu|0\rangle = 0|0\rangle$

$P^\mu(a^\dagger|0\rangle) = a^\dagger(P^\mu + k^\mu)|0\rangle = k^\mu(a^\dagger(k)|0\rangle)$

$P^\mu(a^\dagger(k_1)\cdots a^\dagger(k_n)|0\rangle) = (k_1^\mu + \cdots + k_n^\mu)(\quad\,''\quad)$

結局 $a^\dagger(k)$ は エネルギー-運動量 $k^\mu = (\sqrt{k^2+m^2}, k)$ の粒子 を 1個作る演算子

量子

▷ May 20

$|0\rangle$ ・ $a(k)|0\rangle = 0$ for $\forall k$

$\{a^\dagger(k_1)\cdots a^\dagger(k_n)|0\rangle\}$

↓

P^μ　$k_1^\mu + k_2^\mu + \cdots + k_n^\mu$

Feynmann propagator　　$\theta(t) = \begin{cases}1 & t>0\\ 0 & t<0\end{cases}$

T-product

$T\phi(x)\phi(y) \equiv \theta(x^0-y^0)\,\phi(x)\phi(y) + \theta(y^0-x^0)\,\phi(y)\phi(x)$　　$[a(k), a^\dagger(q)]$

$\langle 0|T\phi(x)\phi(y)|0\rangle = \int\frac{d^3k\,d^3q}{(2\pi)^3\sqrt{2k_0 2q_0}}\Big\{\theta(x^0-y^0)\,e^{-ikx+iqy}\,\underbrace{\langle 0|a(k)a^\dagger(q)|0\rangle}_{\delta^3(k-q)}$

$+\theta(y^0-x^0)\,e^{ikx-iqy}\langle 0|a(q)a^\dagger(k)|0\rangle\Big\}$

$= \int\frac{d^3k}{(2\pi)^3 2k_0}\left\{\theta(x^0-y^0)\,e^{-ik(x-y)} + \theta(y^0-x^0)\,e^{ik(x-y)}\right\}$

$k_0 = \sqrt{k^2+m^2}$

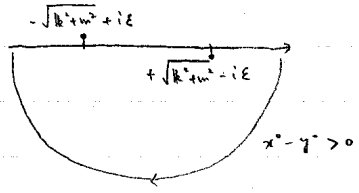
$d^4k = dk^0 d^3k$ $= \int \frac{d^4k}{i(2\pi)^4} \cdot \frac{e^{-ik(x-y)}}{m^2 - k^2 - i\epsilon} \equiv \Delta_F(x-y)$

k_0

$m^2 - (k_0^2 - k^2) - i\epsilon = 0$

$k_0 = \pm\sqrt{k^2+m^2-i\epsilon}$

$= \pm\sqrt{k^2+m^2} \mp i\epsilon$



2-2 複素スカラー場

$\mathcal{L} = \partial_\mu \varphi^\dagger \partial^\mu \varphi - m^2 \varphi^\dagger \varphi = \frac{1}{2}(\partial_\mu \phi_1 \partial^\mu \phi_1 - m^2 \phi_1^2) + \frac{1}{2}(\partial_\mu \phi_2 \partial^\mu \phi_2 - m^2 \phi_2^2)$

$\varphi(x) = \frac{1}{\sqrt{2}}(\phi_1(x) + i\phi_2(x))$

$$\begin{cases} a_\pm(k) = \frac{1}{\sqrt{2}}(a_1(k) \mp i a_2(k)) \\ a_\pm^\dagger(k) = \frac{1}{\sqrt{2}}(a_1^\dagger(k) \pm a_2^\dagger(k)) \end{cases}$$

$$\begin{cases} \varphi(x) \equiv \int \frac{d^3k}{\sqrt{(2\pi)^3 2k_0}} \{ a_-(k) e^{-ikx} + a_+^\dagger(k) e^{ikx} \} \\ \varphi^\dagger(x) \equiv \int \quad " \quad \{ a_+(k) e^{-ikx} + a_-^\dagger(k) e^{ikx} \} \end{cases}$$

位相変換

$$\begin{cases} \varphi(x) \longrightarrow \varphi'(x) = e^{i\theta} \varphi(x) & \delta\varphi = i\theta\varphi \\ \varphi^\dagger(x) \longrightarrow \varphi^{\dagger\prime} = e^{-i\theta} \varphi^\dagger & \delta\varphi^\dagger = -i\theta\varphi^\dagger \end{cases}$$

$\theta \to 1 + i\theta$

の下で $\mathcal{L}$ は不変 $\longrightarrow$ Noether の定理 $\cdots \partial_\mu j^\mu = 0$

$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \cdot i\varphi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi^\dagger)} \cdot (-i\varphi^\dagger)$

$= i(\partial_\mu \varphi^\dagger \cdot \varphi - \varphi^\dagger \partial_\mu \varphi)$

$\downarrow$

$Q = \int d^3x\, j^{\mu=0}(t,x)$

$\frac{dQ}{dt} = 0$

$Q = \int d^3x\, i(\dot{\varphi}\varphi - \varphi^\dagger \dot{\varphi})$

$= \int d^3k\, (a_+^\dagger(k) a_+(k) - a_-^\dagger(k) a_-(k))$

$P^\mu = \int d^3k \sum_{i=1}^{3} k^\mu a_i^\dagger(k) a_i(k)$

$= \int d^3k\, k^\mu (a_+^\dagger(k) a_+(k) + a_-^\dagger(k) a_-(k))$

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ローレンツ群 = 4次元回転 (3,1)

$(\omega, \theta) \to \cosh\theta$ (コンパクト群)

1粒子状態

$a_+^\dagger(k)|0\rangle$, $a_-^\dagger(k)|0\rangle$

$p^\mu \to k^\mu = (\sqrt{k^2+m^2}, k)$ $\quad k^\mu$

$Q \to \quad +1 \quad -1$

$\langle 0|T\varphi(x)\varphi^\dagger(y)|0\rangle = \Delta_F(x-y)$

$\langle 0|T\varphi(x)\varphi(y)|0\rangle = \langle 0|T\varphi^\dagger(x)\varphi^\dagger(y)|0\rangle = 0$



2-3^{-1} SU(N) gauge 場 : Yang-Mills 場

対称性

例 $\mathcal{L} = \partial_\mu\varphi^\dagger\partial^\mu\varphi - m^2\varphi^\dagger\varphi$ $\quad \varphi = \begin{pmatrix}\varphi_1\\ \vdots \\ \varphi_N\end{pmatrix}$

$\varphi \to \varphi' = e^{i\theta}\varphi$

$\varphi_i' = \underbrace{U_i{}^j}_{e^{i\theta}}\varphi_j$ $\quad U$: ユニタリー行列 $\quad T^a$ $a = 0, 1, \cdots, N^2-1$

$G = U(N)$ 群 $\quad \theta$: エルミート行列 $\quad N^2$ 個独立

$\sum_{a=0}^{N^2-1} \theta^a T^a$ $\quad \theta^a$: 実数

$\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$: 物理屋 a convention

・N=2 の場合 $\quad \sigma^1 = \begin{pmatrix}0&1\\1&0\end{pmatrix}$, $\sigma^2 = \begin{pmatrix}0&-i\\i&0\end{pmatrix}$, $\sigma^3 = \begin{pmatrix}1&0\\0&-1\end{pmatrix}$ Pauli 行列

$T^0 = \frac{1}{2}\mathbb{1}_2$

$T^{1\sim3} = \frac{1}{2}\sigma^{1\sim3}$

・N=3

$T^0 = \frac{1}{\sqrt{6}}\mathbb{1}_3$, $T^a = \frac{1}{2}\lambda^a$ $\quad a = 1\sim8$ ← Gell-Mann 行列

$\mathrm{tr}(T^a\mathbb{1}) = 0 \to a = 0, \underbrace{1, 2, \cdots, N^2-1}_{\text{traceless}}$

$\lambda^{1\sim3} = \begin{pmatrix}\boxed{\sigma^{1\sim3}}&0\\0\,0&0\end{pmatrix}$ $\quad \lambda^6 = \begin{pmatrix}0&0&0\\0&0&1\\0&1&0\end{pmatrix}$

$\lambda^4 = \begin{pmatrix}0&0&1\\0&0&0\\1&0&0\end{pmatrix}$ $\quad \lambda^7 = \begin{pmatrix}0&0&0\\0&0&-i\\0&i&0\end{pmatrix}$

$\lambda^5 = \begin{pmatrix}0&0&-i\\0&0&0\\i&0&0\end{pmatrix}$ $\quad \lambda^8 = \frac{1}{\sqrt{3}}\begin{pmatrix}1&0&0\\0&1&0\\0&0&-2\end{pmatrix}$

不変部分群 $H \subset G$

$\forall g \in G \quad gHg^{-1} \subset H \iff H$ は不変部分群

$\forall G$ は自明な不変部分群 : G 自身 , $\{1\}$ 単純群

G が(自明なもの以外に)不変部分群を持たない $\iff$ G : simple

∀ コンパクト群 は (単純群) と (U(1)群) の直積に分解できる

$$U = \exp\left(i\sum_{a=0}^{N^2-1}\theta^a T^a\right) = \underbrace{e^{i\theta^0 T^0}}_{e^{i\theta^0}} \cdot \underbrace{e^{i\sum_{a=1}^{N^2-1}\theta^a T^a}}_{\det = 1} \qquad \{e^{i\theta^0 \mathbb{1}}\} = U(1)\text{群}$$

$U(N) = U(1) \times SU(N)$

↳ $\det U = 1$ である

$N\times N$ ユニタリー行列全体

$$SU(N) = \left\{ e^{i\theta^a T^a} \;\middle|\; \mathrm{tr}\, T^a = 0,\ a = 1, 2, \cdots, N^2-1 \right\}$$

↑ 実パラメータ　　$\dim SU(N) = N^2 - 1$

generator T^a　　$SU(2) \simeq SO(3)$

$[T^a, T^b] = i f_{abc} T^c$　　f_{abc} : 群の構造定数

$N=2 \to f_{abc} = \epsilon_{abc}$

表現

2×2　$\rho_{\frac{1}{2}}(J_a) = \frac{1}{2}\sigma_a$　　$j = \frac{1}{2}$　　$\rho_{\frac{1}{2}}(J_3) = \frac{1}{2}\sigma_3 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}$

3×3　$[\rho_1(J_a)]_{bc} = i\epsilon_{bac}$　　$j = 1$　　$\rho_1(J_3) = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

↓

$\begin{pmatrix} +1 & & \\ & 0 & \\ & & -1 \end{pmatrix}$ $\begin{matrix} + \\ 0 \\ - \end{matrix}$

J_3 の固有値は量子化されている

$\frac{1}{2}\times$ 整数倍

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局所ゲージ変換

$$\mathcal{L} = \partial_\mu \varphi^\dagger \partial^\mu \varphi - m^2 \varphi^\dagger \varphi$$

$$\varphi = \begin{pmatrix} \varphi_1 \\ \vdots \\ \varphi_N \end{pmatrix}$$

$1 + i\theta^a T^a + O(\theta^2)$

$U(1)$

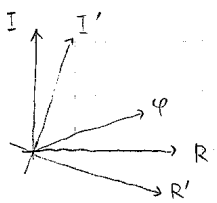
$\varphi \to \varphi' = U\varphi \qquad U \in SU(N)$

$\varphi' = e^{i\theta}\varphi$　　$\varphi'_i = U_i{}^j \varphi_j$　　実 ↓ ↙ generator 行列　　構造定数 ↓

$R + iI$　　$U_i{}^j = [\exp(i\theta^a T^a)]_i{}^j$　　$[T^a, T^b] = i f_{abc} T^c$

$\begin{pmatrix} R \\ I \end{pmatrix}' = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} R \\ I \end{pmatrix}$

$a, b, c = 1 \sim N^2-1$



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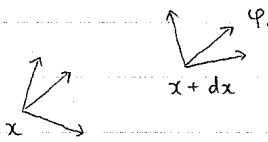
「時空の各点各点で勝手に座標軸を設定しても、理論の形は同じであるべき」

—— ゲージ原理

$\varphi(x) \longrightarrow \varphi'(x) = U(x)\varphi(x)$ $\qquad U(x) = \exp[ig\theta^a(x)T^a]$

$U(x)$ ∈ $SU(N)$

"平行移動"の定義



xに於けるベクトル $\varphi(x)$ を $x+dx$ の点に

平行移動したもの $\varphi_{//}(x+dx) = \varphi(x) + igA^a_\mu (T^a)_i{}^j \varphi_j(x)\, dx^\mu$

$A^a_\mu(x)$: 接続場 (connection)

ゲージ場

$tr(T^aT^b) = \eta\,\delta^{ab}$

matrix 記法

$\sum_{a=1}^{N^2-1} A^a_\mu(x)(T^a)_i{}^j \equiv A_\mu(x) \longrightarrow A^a_\mu = \eta^{-1}\, tr(T^a A_\mu)$

$\varphi_{//}(x+dx)$ は $x+dx$ に於ける量 : $\varphi'_{//}(x+dx) = U(x+dx)\,\varphi_{//}(x+dx)$

差 : $\varphi(x+dx) - \varphi_{//}(x+dx)$

$= \varphi(x) + \partial_\mu\varphi(x)\,dx^\mu - (\varphi(x) + igA_\mu(x)\varphi(x)\,dx^\mu)$

$= (\partial_\mu - igA_\mu(x))\,\varphi(x)\,dx^\mu$

$= D_\mu\varphi(x)\,dx^\mu$

共変微分 (covariant derivative)

$\varphi'(x+dx) - \varphi'_{//}(x+dx)$

$= U(x+dx)\,(\varphi(x+dx) - \varphi_{//}(x+dx))$

$\downarrow$ $\qquad\qquad\qquad

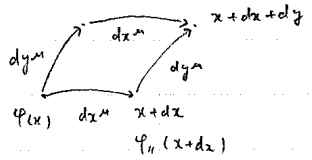
ゲージ場の変換則

局所ゲージ変換

$$A_\mu(x) \longrightarrow A'_\mu(x) = \frac{i}{g} U(x)\,\partial_\mu U^{-1}(x) + U(x) A_\mu(x) U^{-1}(x) \qquad \textcircled{2}$$

$$\varphi \to \varphi'(x) = U(x)\varphi(x)$$

場の強さ（曲率）



$$\varphi_\mu(x+dx) = \varphi(x+dx) - D_\mu \varphi(x)\,dx^\mu$$

$$\varphi_{\mu\nu}''(x+dx+dy) = \varphi_\mu(x+dx+dy) - D_\nu \varphi_\mu(x+dx)\,dy^\mu$$

$$= \varphi(x+dx+dy) - D_\mu \varphi(x+dy)\,dx^\mu - D_\nu \varphi(x+dx)\,dy^\mu + D_\nu D_\mu \varphi(x)\,dx^\mu dx^\nu$$

$$-) \quad \varphi_{\nu\mu}''(x+dx+dy) = (dx \leftrightarrow dy)$$

$$\Delta\varphi(x+dx+dy) = (D_\nu D_\mu - D_\mu D_\nu)\varphi\,dx^\mu dx^\nu$$

$$= -[D_\mu, D_\nu]\varphi\,dx^\mu dx^\nu$$

$-igF_{\mu\nu}(x) \equiv [D_\mu, D_\nu]$ ：微分がなくなっている

$$F_{\mu\nu}(x) = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]$$

$\longrightarrow$ 場の強さ（曲率）

$$[\partial_\mu - igA_\mu, \partial_\nu - igA_\nu]$$

$$= (\partial_\mu A_\nu) + ig(\partial_\nu A_\mu) + (-ig)^2 [A_\mu, A_\nu]$$

$$D_\mu \to D'_\mu = U(x) D_\mu U^{-1}(x)$$

$$F_{\mu\nu} \to F'_{\mu\nu}(x) = U(x) F_{\mu\nu}(x) U^{-1}(x)$$

$\longrightarrow$ 場の強さの変換則

$$\epsilon^{\mu\nu\rho\sigma}[D_\nu, [D_\rho, D_\sigma]] = 0 \quad \text{Jacobi identity}$$

$$\epsilon^{\mu\nu\rho\sigma}\{\partial_\nu F_{\rho\sigma} - ig[A_\nu, F_{\rho\sigma}]\} = 0 \quad \text{Bianchi 恒等式} \qquad \textcircled{1}$$

成分表示 $F_{\mu\nu}(x) = \sum F^a_{\mu\nu}(x)\,T^a$

$F^a_{\mu\nu}(x)$ ：成分　　SU(N)：N^2-1 成分

$$A_\mu(x) = \sum A^a_\mu(x)\,T^a$$

$$F^a_{\mu\nu}(x) = \partial_\mu A^a_\nu(x) - \partial_\nu A^a_\mu(x) + g f_{abc} A^b_\mu(x) A^c_\nu(x)$$

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無限小ゲージ変換 : $U(\theta(x)) \xrightarrow{\theta(x)\ll 1} 1 + ig\theta^a(x)T^a$

②

$$\varphi'(x) - \varphi(x) = \begin{cases} \delta\varphi(x) = ig\theta^a(x)T^a\varphi(x) \\ \delta A^a_\mu(x) = \partial_\mu\theta^a(x) + g f_{abc} A^b_\mu(x)\theta^c(x) \end{cases}$$

問1. ①,②を導け

ゲージ不変な Lagrangian

$\psi \to U\psi$

$$\mathcal{L} = \mathcal{L}_{\text{gauge場}} + \mathcal{L}_{\text{matter場}} \qquad \left(\mathcal{L}_{\text{Dirac場}} = \bar\psi(i\gamma^\mu\partial_\mu - m)\psi \right)$$

$$\mathcal{L}_{\text{gauge場}} = -\frac{1}{4}\cdot\frac{1}{n}\,\mathrm{tr}\left(F_{\mu\nu}(x)F^{\mu\nu}(x)\right) = -\frac{1}{4}F^a_{\mu\nu}(x)F^{a\mu\nu}(x)$$

$$\left((D_\mu\varphi)' = U(x)\,D_\mu\varphi \quad \varphi' = U(x)\,\varphi(x) \right)$$

$$\mathcal{L}_{\text{matter場}} = \begin{cases} (D_\mu\varphi)^\dagger D^\mu\varphi - m^2\varphi^\dagger\varphi \\ \bar\psi(i\gamma^\mu D_\mu - m)\psi \end{cases}$$

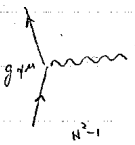
$\partial_\mu - igA_\mu$

$$\bar\psi(i\gamma^\mu D''_\mu - m)\psi = \bar\psi(i\gamma^\mu\partial_\mu - m)\psi + g\bar\psi\gamma^\mu A_\mu\psi$$

相互作用

$g\bar\psi\gamma^\mu\psi)\cdot A_\mu$

current j^μ ↑ 電磁場



$$A_\mu = \sum_{a=1}^{N^2-1} A^a_\mu(x)T^a$$

$$(D_\mu\varphi)^\dagger(D^\mu\varphi) - m^2\varphi^\dagger\varphi = (\partial_\mu\varphi^\dagger\partial^\mu\varphi - m^2\varphi^\dagger\varphi) - ig(\partial_\mu\varphi^\dagger A_\mu\varphi - \varphi^\dagger A_\mu\partial_\mu\varphi) + g^2\varphi^\dagger A_\mu A^\mu\varphi$$

問2. U(1) のとき $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + j^\mu A_\mu$ $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$: U(1)

↳ Euler-Lagrange eq

$$\begin{cases} \partial^\nu F_{\mu\nu} = j_\mu \cdots \text{E-L} \\ \epsilon^{\mu\nu\rho\sigma}\partial_\nu F_{\rho\sigma} = 0 \cdots \text{Bianchi} \end{cases}$$

……真空中の Maxwell 方程式と一致していることを示せ

2-3 (massless) ベクトル場 (とくに電磁場)

$\delta A^a_\mu = \partial_\mu \theta^a$ ┘ $+ g f_{abc} A^b_\mu \theta^c$

$\mathcal{L} = -\frac{1}{4}(F^a_{\mu\nu} F^{a\mu\nu})$ $\quad F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f_{abc} A^b_\mu A^c_\nu$

$= -\frac{1}{4}(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)^2$ ┘ + (相互作用項)

↳ $\delta A^a_\mu = \partial_\mu \theta^a$

$\dot{\phi} = \frac{\partial}{\partial t}\phi$

$\pi_0 \equiv \frac{\partial L}{\partial \dot{A}_0} = 0$ 特異系

$\dot{A}_0$ が現われない

量子化できない $\quad [A_0(x), \pi_0(y)] = i\delta^3(x-y)$

$\Rightarrow -\frac{1}{4}(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)^2 - \underbrace{\frac{1}{2\alpha}(\partial^\mu A^a_\mu)^2}$ → Fermi 項

gauge parameter

$\alpha = 1$ Feymann ゲージ

$\alpha = 0$ Landau ゲージ

$= \underbrace{-\frac{1}{2}(\partial_\mu A_\nu)^2} + \frac{1}{2}(\partial^\mu A_\mu)^2 - \frac{1}{2\alpha}(\partial^\mu A_\mu)^2$

$\alpha = 1$

$= -\frac{1}{2}\partial_\mu A_0 \partial^\mu A_0 + \frac{1}{2}\partial_\mu A_1 \partial^\mu A_1 + \frac{1}{2}\partial_\mu A_2 \partial^\mu A_2 + \frac{1}{2}\partial_\mu A_3 \partial^\mu A_3$

$\Box A_\mu = 0$ $\qquad A^a_\mu(x) = \int \frac{d^3k}{\sqrt{2k_0(2\pi)^3}} \left(a^a_\mu(k) e^{-ikx} + a^{a\dagger}_\mu(k) e^{ikx} \right)$

$k_0 = \sqrt{k^2}$

CCR → $[a^a_\mu(k), a^{b\dagger}_\nu(\ell)] = -g_{\mu\nu}\delta^{ab}\delta^3(k-\ell)$ $\qquad g_{\mu\nu} = \begin{pmatrix} +1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$

正準交換関係
canonical comitate relation

$\begin{pmatrix} A_0 \\ A_1 \\ A_2 \\ A_3 \end{pmatrix} = \int \left[\begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{pmatrix} e^{-ikx} + \text{h.c.} \right]$ (h.c.: hermite conj)

$a_0 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + a_1 \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + a_3 \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \sum_{\sigma = \pm, L, S} a(k,\sigma)\, \varepsilon_\mu(k,\sigma)$

base のとりかえ

$k \parallel e_3$ $\qquad \varepsilon^\mu(k, \sigma = \pm) \equiv \begin{pmatrix} 0 \\ 1 \\ \pm i \\ 0 \end{pmatrix} / \sqrt{2}$ $\qquad \varepsilon^\mu(k, L) \equiv \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ 縦波 $\qquad \varepsilon^\mu(k, S) \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

横波偏極ベクトル

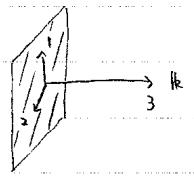
QED

$G = U(1)$

$$A_\mu(x) = \int \frac{d^3k}{\sqrt{2k_0(2\pi)^3}} \sum_{\sigma=\pm,L,S} \left(a(k,\sigma)\,\varepsilon_\mu(k,\sigma)\,e^{-ikx} + \text{h.c.} \right)$$

$k \parallel e_3$ のとき

$$\begin{cases} a(k,\pm) = (a^1(k) \mp i a^2(k))/\sqrt{2} & \cdots \text{横波} \\ a(k,L) = a^3(k) \\ a(k,S) = a^0(k) \end{cases} \Big\} \text{非物理的}$$



$$[a(k,\sigma), a^\dagger(q,\tau)] = \delta_{\sigma\tau}\,\delta^3(k-q)$$

$$[a(k,L), a^\dagger(q,L)] = \delta^3(k-q) \qquad [a, a^\dagger] = -1$$

$$[a(k,S), a^\dagger(q,S)] = -\delta^3(k-q) \quad (\text{困る！})$$

$$\| a^\dagger(0)|0\rangle \|^2 = \langle 0 | \underbrace{a a^\dagger}_{a^\dagger a - 1} |0\rangle = -1 \underbrace{\langle 0|0\rangle}_{1} < 0$$

Gupta-Bleuler 補助条件

$$\langle \text{phys}' | (\partial^\mu A_\mu) | \text{phys} \rangle = 0$$

$$\Leftarrow (\partial^\mu A_\mu)^{(+)}(x) |\text{phys}\rangle = 0$$

消滅演算子部分

$$k^\mu = \begin{pmatrix} k \\ 0 \\ 0 \\ k \end{pmatrix}$$

$$\to B^{(+)}(x)|\text{phys}\rangle = 0$$

中西-Lautrup 補助条件

$$(a(k,S) - a(k,L))|\text{phys}\rangle = 0 \to \underbrace{a^\dagger(k,S) + a^\dagger(k,L)}$$

のモードを禁止

$$|\text{phys}\rangle = \underbrace{a^\dagger(k_1,\pm)\cdots a^\dagger(k_N,\pm)}_{\text{横波モード}} \left(1 + 0\,\underbrace{b^\dagger(q)}_{\text{見えない}} + 0\,\underbrace{b^\dagger(q_1)b^\dagger(q_2)}_{\text{見えない}} + \cdots \right)|0\rangle$$

$$b(k) \equiv a(k,S) - a(k,L)$$

$$[a(S) - a(L), a^\dagger(S) - a^\dagger(L)] = -1 + 1 = 0$$

$$\| b^\dagger(k)|0\rangle \|^2 = \langle 0 | b(k) \underbrace{b^\dagger(k)|0\rangle}_{} = 0$$

$$P^\mu = \int d^3k\; k^\mu \left(-a^\dagger_\nu(k) a^\nu(k) \right)$$

$$= \int d^3k\; k^\mu \left[+a^\dagger(k,+)a(k,+) + a^\dagger(k,-)a(k,-) + a^\dagger(k,L)a(k,L) \underbrace{- a^\dagger(k,S)a(k,S)}_{} \right]$$

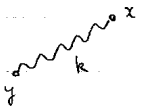
$$P^\mu\, a^\dagger(k,\pm)|0\rangle = k^\mu\, a^\dagger(k,\pm)|0\rangle$$

$$J_3\; a^\dagger(k,\pm)|0\rangle = \pm 1\, a^\dagger(k,\pm)|0\rangle$$

↳ photon はスピン 1 を持つ

propagator

$$\langle 0|T A_\mu(x) A_\nu(y)|0\rangle = \int \frac{d^4k}{i(2\pi)^4} e^{-ik(x-y)} \boxed{\frac{g_{\mu\nu}}{k^2}}$$



2-4 スピノール場 (Dirac場) $\to \psi = \begin{pmatrix}\psi_1\\ \psi_2\\ \psi_3\\ \psi_4\end{pmatrix}$ ← 標準表示 (Dirac表示)

$\mathcal{L} = \bar\psi(i\gamma^\mu\partial_\mu - m)\psi$ $\quad \gamma^0 = \begin{pmatrix}\mathbb{1} & 0\\ 0 & -\mathbb{1}\end{pmatrix}$ $\quad \vec\gamma = \begin{pmatrix}0 & \vec\sigma\\ -\vec\sigma & 0\end{pmatrix}$

$\pi^\alpha_\psi = \frac{\partial\mathcal{L}}{\partial\dot\psi_\alpha} = (\bar\psi i\gamma^0)_\alpha = i\psi^\dagger_\alpha$ $\quad \gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = 2g_{\mu\nu}$

$\bar\psi \equiv \psi^\dagger\gamma_0$ (Dirac共役)

正準反交換関係 $\{\psi_\alpha(x,t), \pi^\beta_\psi(y,t)\} = i\delta^\beta_\alpha\delta^3(x-y)$

(CACR) $\hookrightarrow \{\psi_\alpha(x,t), \psi^{\beta\dagger}(y,t)\} = \delta^\beta_\alpha\delta^3(x-y)$

$H = \int d^3x(\pi_\psi\dot\psi - \mathcal{L}) = \int d^3x\,\bar\psi(-i\vec\gamma\cdot\nabla + m)\psi$

$i\dot\psi = [\psi, H]$ $\quad \boxed{(i\gamma^\mu\partial_\mu - m)\psi = 0}$

Dirac方程式

$$\boxed{\psi(x) = \int \frac{d^3p}{\sqrt{(2\pi)^3 2p_0}} \sum_{s=\pm1} \left(b(p,s)u(p,s)e^{-ipx} + d^\dagger(p,s)v(p,s)e^{ipx}\right)}$$

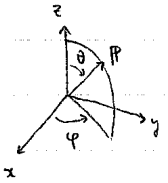
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$\not p \equiv p_\mu\gamma^\mu$ $\quad (\not p - m)u(p,s) = 0$ (4×4)

$\gamma^0 = \begin{pmatrix}1 & \\ & -1\end{pmatrix}$ $\quad \vec\gamma = \begin{pmatrix} & \vec\sigma\\ -\vec\sigma & \end{pmatrix}$ $\quad \not p = \begin{pmatrix}p_0 - m & -\vec p\cdot\vec\sigma\\ \vec p\cdot\vec\sigma & -p_0 - m\end{pmatrix}$

$u(p,s) = \begin{pmatrix} w_s \\ \frac{\vec p\cdot\vec\sigma}{p_0+m} w_s\end{pmatrix}$ $\quad \sigma_z$の固有状態 $\begin{cases} w_s = \begin{pmatrix}1\\0\end{pmatrix}\\ w_s = \begin{pmatrix}0\\1\end{pmatrix}\end{cases}$

$\vec\sigma\cdot\vec p/|\vec p| = h$ $\quad$ helicityの固有状態 $\begin{cases}\chi_{h=+1} = \begin{pmatrix} e^{-i\frac{\varphi}{2}}\cos\frac{\theta}{2}\\ e^{i\frac{\varphi}{2}}\sin\frac{\theta}{2}\end{pmatrix}\\ \chi_{h=-1} = \begin{pmatrix} e^{-i\frac{\varphi}{2}}\sin\frac{\theta}{2}\\ e^{i\frac{\varphi}{2}}\cos\frac{\theta}{2}\end{pmatrix}\end{cases}$



$(-\not p - m)v(p,s) = 0$ $\quad v(p,s) = i\gamma^2 u^*(p,s)$

$\{b,b\} = \{d,d\} = \{b,d^\dagger\} = 0$

$\begin{cases}\{b(p,s), b^\dagger(q,t)\} = \delta_{st}\delta^3(\vec p - \vec q)\\ \{d(p,s), d^\dagger(q,t)\} = \ \ 〃\end{cases}$

$\alpha = \gamma_0\vec\gamma$ $\quad \beta = \gamma_0$

$p_0 u(p,s) = (\vec p\cdot\vec\alpha + m\beta)u(p,s)$

$-p_0 v(p,s) = (\vec p\cdot\vec\alpha + m\beta)v(p,s)$

$H = \vec\alpha\cdot\vec p + \beta m$

規格直交性

$$\Rightarrow \begin{cases} \bar{u}(\mathbf{p},s)\,u(\mathbf{p},t) = 2m\,\delta_{st} \\ \bar{v}(\mathbf{p},s)\,v(\mathbf{p},t) = -2m\,\delta_{st} \\ u^{\dagger}(\mathbf{p},s)\,v(-\mathbf{p},t) = v^{\dagger}(\mathbf{p},s)\,u(-\mathbf{p},s) = 0 \\ \bar{u}(\mathbf{p},s)\,\gamma^{\mu}\,u(\mathbf{p},t) = \bar{v}(\mathbf{p},s)\,\gamma^{\mu}\,v(\mathbf{p},t) = 2p^{\mu}\delta_{st} \end{cases}$$

完全性

$$\begin{cases} \sum_{s=\pm 1} u(\mathbf{p},s)\,\bar{u}(\mathbf{p},s) = \not{p} + m \\ \sum_{s=\pm 1} v(\mathbf{p},s)\,\bar{v}(\mathbf{p},s) = \not{p} - m \end{cases}$$

エネルギー・運動量演算子　　$\mathcal{L} = \bar{\psi}(i\gamma^{\mu}\partial_{\mu} - m)\psi$

$$P^{\mu} = \int d^3x \left(\underbrace{\frac{\partial \mathcal{L}}{\partial \dot{\psi}}}_{\bar{\psi} i\gamma^0} \partial^{\mu}\psi - g^{\mu 0}\mathcal{L} \right) \qquad \underline{p^{\mu} = (\sqrt{p^2+m^2}, \mathbf{p})}$$

$$= \int d^3\mathbf{p}\; p^{\mu} \left(b^{\dagger}(\mathbf{p},s)\,b(\mathbf{p},s) \underbrace{- d(\mathbf{p},s)\,d^{\dagger}(\mathbf{p},s)}_{+\,d^{\dagger}(\mathbf{p},s)\,d(\mathbf{p},s)} \underbrace{- \delta^3(0)}_{V/(2\pi)^3} \right)$$

角運動量演算子

$$\mathbb{J} = \int d^3x\; \psi^{\dagger}(\mathbf{x},t) \left(\mathbb{L} + \frac{1}{2}\boldsymbol{\sigma} \right) \psi(\mathbf{x},t) \qquad \mathbb{L} = \mathbf{x} \times \mathbf{p} \;(\to -i\nabla)$$

$$= \int d^3p \sum_s \left(b^{\dagger}(\mathbf{p},s)\,\mathbb{L}\,b(\mathbf{p},s) + d^{\dagger}(\mathbf{p},s)\,\mathbb{L}\,d(\mathbf{p},s) \right) + \sum_{s,s'} \left(b^{\dagger}(\mathbf{p},s) \left(\frac{\boldsymbol{\sigma}}{2}\right)_{ss'} b(\mathbf{p},s') + d^{\dagger}(\mathbf{p},s) \left(\frac{\boldsymbol{\sigma}}{2}\right)_{ss'} d(\mathbf{p},s') \right)$$

$b(\mathbf{p},s)|0\rangle = d(\mathbf{p},s)|0\rangle = 0$

$P^{\mu}\,\underbrace{b^{\dagger}(\mathbf{p},s)}_{d^{\dagger}}|0\rangle = p^{\mu}\,\underbrace{b^{\dagger}(\mathbf{p},s)}_{d^{\dagger}}|0\rangle$

$\mathbb{J}\,b^{\dagger}(\mathbf{p}=0,s)|0\rangle = \left(\frac{\boldsymbol{\sigma}}{2}\right)_{ss'} b^{\dagger}(\mathbf{p},s')|0\rangle$

↳ $j = \frac{1}{2}$, $s = \pm 1 \;\to\; j_z = \pm\frac{1}{2}$

電荷；$\psi \to e^{i\frac{e}{\hbar}\theta}\psi$, $\bar{\psi} \to e^{-i\frac{e}{\hbar}\theta}\bar{\psi}$

$$j^{\mu} = \frac{\partial \mathcal{L}}{\partial(\partial_{\mu}\psi)}\,i\psi + \frac{\partial \mathcal{L}}{\partial(\partial_{\mu}\bar{\psi})}(-i\bar{\psi}) \qquad Q = \int d^3x\,(-e)\,\psi^{\dagger}\psi$$

$$= \underbrace{-\bar{\psi}\gamma^{\mu}\psi}_{\hat{e}} \qquad = \int d^3\mathbf{p}\,(-e)\left[b^{\dagger}(\mathbf{p},s)\,b(\mathbf{p},s) - d^{\dagger}(\mathbf{p},s)\,d(\mathbf{p},s) \right]$$

$$\begin{cases} Q\, b^\dagger(\mathbf{p},s)|0\rangle = -e\, b^\dagger(\mathbf{p},s)|0\rangle \\ Q\, d^\dagger(\mathbf{p},s)|0\rangle = +e\, d^\dagger(\mathbf{p},s)|0\rangle \end{cases}$$

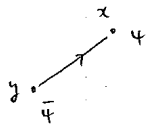
$b^\dagger$ と $d^\dagger$ の粒子は逆符号の電荷 (反粒子)

$$\begin{cases} \{\psi,\psi\}=0 \\ \{\psi,\bar\psi\}=\delta^3(x-y) \end{cases}$$

Feynman propagator

$$T\psi(x)\bar\psi(y) = \theta(x^0-y^0)\psi(x)\bar\psi(y) - \theta(y^0-x^0)\bar\psi(y)\psi(x)$$

$$\langle 0|T\psi(x)\bar\psi(y)|0\rangle = \int \frac{d^3\mathbf{p}}{(2\pi)^3 2p_0}\left\{\theta(x^0-y^0)\,e^{-ip(x-y)}(\not p+m) + \theta(y^0-x^0)\,e^{ip(y-x)}(-\not p+m)\right\}$$



$$= \int \frac{d^4p}{i(2\pi)^4}\, e^{-ip(x-y)} \boxed{\frac{m+\not p}{m^2-p^2-i\epsilon}} = S_f(x-y)$$

$$\frac{1}{m-\not p-i\epsilon}$$

3 相互作用系：摂動論

例 QED (quantum electro-dynamics)　ψ：電子の場　A_μ：電磁場

$$\mathcal{L} = \underbrace{\bar\psi(i\gamma^\mu\partial_\mu - m)\psi - \frac{1}{2}\partial_\mu A_\nu \partial^\mu A^\nu}_{\mathcal{L}_{free}} + \underbrace{e\bar\psi\gamma^\mu\psi\cdot A_\mu}_{\mathcal{L}_{int}} \qquad U(1)$$

$$L = \int d^3x\,\mathcal{L} \quad\Rightarrow\quad H = H_{free} + H_{int}$$

$$H_{int} = -\int d^3x\,\mathcal{L}_{int} = -e\int d^3x\,\bar\psi\gamma_\mu\psi\cdot A^\mu(x)$$

$$H_{free} = \int d^3x\left[\bar\psi(i\gamma\nabla+m)\psi - \frac{1}{2}(\pi_\mu\pi^\mu + \nabla A_\mu\nabla A^\mu)\right]$$

3. 相互作用系、摂動論

例

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - \frac{1}{2}\partial_\mu A_\nu \partial^\mu A^\nu + e\bar{\psi}\gamma_\mu\psi A^\mu$$

$$H = H_{free} + H_{int}$$

$$H_{free} = \int d^3x \left[\bar{\psi}(\mathbf{x})(-i\boldsymbol{\alpha}\cdot\nabla + m)\psi(\mathbf{x}) - \frac{1}{2}(\pi_\mu(\mathbf{x})\pi^\mu(\mathbf{x}) + \nabla A_\mu(\mathbf{x})\cdot\nabla A^\mu(\mathbf{x})\right]$$

$$H_{int} = \int d^3x \left[-e\bar{\psi}(\mathbf{x})\gamma_\mu\psi(\mathbf{x})A^\mu(\mathbf{x})\right]$$

Schrödinger 表示（状態が時間発展）　op. は動かない

$$i\frac{\partial}{\partial t}|\Psi_S(t)\rangle = H|\Psi_S(t)\rangle \qquad |\Psi_S(t)\rangle = e^{-iHt}\underbrace{|\Psi_S(t=0)\rangle}_{=|\Psi_H\rangle}$$

Heisenberg 表示（状態は動かず、op が時間発展）

$$\begin{cases}\psi(\mathbf{x},t) = e^{iHt}\psi(\mathbf{x})e^{-iHt}\\ A_\mu(\mathbf{x},t) = e^{iHt}A_\mu(\mathbf{x})e^{-iHt}\end{cases} \longleftrightarrow \begin{cases}\dot{\psi} = i[H,\psi]\\ \dot{A} = i[H,A]\end{cases}$$

$$|\Psi_H\rangle = |\Psi_S(t=0)\rangle$$

相互作用表示（状態は H_{int} で発展、演算子は H_{free} で発展）

$$\text{演算子}\begin{cases}\psi(\mathbf{x},t) = e^{iH_{free}t}\psi(\mathbf{x})e^{-iH_{free}t}\\ \bar{\psi}(\mathbf{x},t) = \text{〃}\quad\bar{\psi}(\mathbf{x})\quad\text{〃}\\ A^\mu(\mathbf{x},t) = \text{〃}\quad A^\mu(\mathbf{x})\quad\text{〃}\end{cases}$$

朝永 - Schwinger + Feynmann

$$\langle\Psi_S'(t)|F(\psi(\mathbf{x}),\bar{\psi}(\mathbf{y}),A_\mu(\mathbf{z}))|\Psi_S(t)\rangle$$

$$\langle\Psi_H'|e^{iHt}\quad e^{-iHt}|\Psi_H\rangle$$

$$= \langle\Psi_H'|F(\psi(\mathbf{x},t),\bar{\psi}(\mathbf{y},t),A_\mu(\mathbf{z},t))|\Psi_H\rangle$$

$$|\Psi_I(t)\rangle = e^{iH_{free}t}|\Psi_S(t)\rangle$$

$$i\frac{\partial}{\partial t}|\Psi_I(t)\rangle = -H_{free}|\Psi_I(t)\rangle + e^{iH_{free}t}(H_{free}+H_{int})e^{-iH_{free}t}|\Psi_I(t)\rangle$$

$$= H_{int}(t)|\Psi_I(t)\rangle$$

$$H_{int}(t) = e^{iH_{free}t}H_{int}e^{-iH_{free}t} = \int d^3x\left[-e\bar{\psi}(\mathbf{x},t)\gamma_\mu\psi(\mathbf{x},t)A^\mu(\mathbf{x},t)\right]$$

$\dot{\psi} = [H_{free},\psi] \rightarrow (i\gamma^\mu\partial_\mu - m)\psi = 0 \rightarrow \psi(\mathbf{x},t) = \int\frac{d^3\mathbf{p}}{\sqrt{2p_0(2\pi)^3}}\left(\sum_s b(\mathbf{p},s)u(\mathbf{p},s)e^{-ipx} + d^\dagger(\mathbf{p},s)v(\mathbf{p},s)e^{ipx}\right)$　（$e^{-ip_0t+i\mathbf{p}\cdot\mathbf{x}}$）

$\dot{A} = [H_{free},A] \rightarrow \Box A_\mu = 0$

$$i\frac{\partial}{\partial t}|\Psi_I(t)\rangle = H_{int}(t)\,|\Psi_I(t)\rangle$$

$$|\Psi_I(t)\rangle = U(t;t_0)\,|\Psi_I(t_0)\rangle \qquad U(t;t_0) = e^{iH_{free}t}\,e^{-iH(t-t_0)}\,e^{-iH_{free}t_0}$$

$$\underline{U(t;t_0) = T\exp\left[-i\int_{t_0}^{t} dt'\,H_{int}(t')\right]}$$

$$\underline{i\frac{\partial}{\partial t}U(t;t_0) = H_{int}(t)\,U(t;t_0)} \qquad U(t_0;t_0)=1$$

$$U(t;t_0) = 1 - i\int_{t_0}^{t} dt'\,H_{int}(t')\,\underline{U(t';t_0)}$$

$$= 1 - i\int_{t_0}^{t} dt'\,H_{int}(t') + (-i)^2\int_{t_0}^{t} dt'\int_{t_0}^{t'} dt''\,H_{int}(t')H_{int}(t'') + \cdots$$

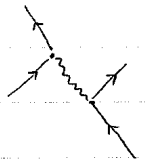
$$\downarrow \quad t' > t''$$

$$\frac{(-i)^2}{2!}\int_{t_0}^{t} dt'\int_{t_0}^{t} dt''\;T\left(H_{int}(t')\,H_{int}(t'')\right)$$

Dyson の S 行列

$$\boxed{S = U(+\infty;-\infty) = T\exp\left[-i\int d^4x\,\mathcal{H}_{int}(x)\right]}$$

$$\mathcal{H}_{int} = -e\,\bar{\Psi}(x)\,\gamma^\mu\,\Psi(x)\,A_\mu(x)$$



散乱振幅 $\qquad S = 1 + i\hat{T}$

$$\langle\beta|S|\alpha\rangle = \langle\beta|\alpha\rangle + i(2\pi)^4\delta^4(p_\beta - p_\alpha)\underbrace{\boxed{\langle\beta|T|\alpha\rangle}}_{a_{m_{\beta\alpha}}}$$

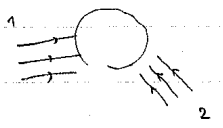
例) 2電子状態

$$|\alpha\rangle = \frac{b^\dagger(\mathbf{p},s)}{\sqrt{2p_0(2\pi)^3}}\frac{b^\dagger(\mathbf{p}',s')}{\sqrt{2p_0'(2\pi)^3}}|0\rangle \qquad |\mathbf{p}\rangle = \sqrt{(2\pi)^3 2p_0}\;b(\mathbf{p},s)|0\rangle$$

$$\langle\mathbf{p}|\mathbf{p}'\rangle = (2\pi)^3\,2p_0\,\delta^3(\mathbf{p}-\mathbf{p}')$$

散乱断面積 cross section

$$\boxed{1+2 \to (3+4+\cdots+n)}$$



$$p_i^\mu = (p_i^0 = \sqrt{\mathbf{p}_i^2 + m_i^2},\ \mathbf{p}_i)$$

粒子1の個数密度 ρ_1

〃 2 〃 ρ_2

No. 24

Date 94 · 6 · 24

来週休講

粒子1と2の"相対速度" $\bar{v}_{rel} = \dfrac{\sqrt{(P_1\cdot P_2)^2 - m_1^2 m_2^2}}{P_1^0 P_2^0}$

$a\mathbf{v}_1 + b\mathbf{v}_2 = 0$ が成立するLorentz系では

$\bar{v}_{rel} = |\mathbf{v}_1 - \mathbf{v}_2|$ が成立

時間 T, 体積 V の中で N 回起こった

cross section σ

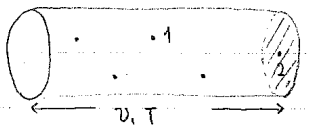
$N = \rho_1\rho_2 \bar{v}_{rel} VT\sigma$

粒子の流れ $j^\mu = (\rho, \rho\mathbf{v}) = \rho(1, \mathbf{v})$

$p^\mu = (p^0, \mathbf{p}) = p^0(1, \mathbf{v})$

$\mathbf{p}/p^0 = $ scalor

粒子2の静止系



$\rho_1(v_1 T\sigma) \times \rho_2 V = N$

$\langle\beta|S|\alpha\rangle = \langle\beta|\alpha\rangle + i(2\pi)^4\delta^4(P_\beta - P_\alpha)\, m_{\alpha\beta}$

$m_{\alpha\beta} = \langle\beta|T|\alpha\rangle$

$\langle\mathbf{p}|\mathbf{p}'\rangle = (2\pi)^3 2p_0\,\delta^3(\mathbf{p}-\mathbf{p}') \longrightarrow \boxed{\int \frac{d^3\mathbf{p}}{(2\pi)^3 2p_0}|\mathbf{p}\rangle\langle\mathbf{p}| = 1}$

$(2\pi)^3\int d^3x\, e^{i\mathbf{0}\cdot\mathbf{x}} = V$

$\langle\mathbf{p}|\mathbf{p}\rangle = 2p_0\left((2\pi)^3\delta^3(0)\right)$

$= 2p_0 V$

$|\alpha\rangle = |\mathbf{p}_1, \mathbf{p}_2\rangle \longrightarrow S|\alpha\rangle$

$|\beta\rangle = |\mathbf{p}_3, \mathbf{p}_4 \cdots\rangle$

$\underbrace{\langle\beta|S|\alpha\rangle}_{i(2\pi)^4\delta^4(P_\alpha - P_\beta)\, m_{\alpha\beta}}$

$(\mathbf{p}_j, \mathbf{p}_j + d\mathbf{p}_j)$

$dN = \prod_{j=3}^{n}\frac{d^3\mathbf{p}_j}{(2\pi)^3 2p_j^0}\,|\langle\beta|S|\alpha\rangle|^2 \xrightarrow{\times\frac{1}{\rho_1\rho_2\bar{v}_{rel}VT}} \boxed{d\sigma = \prod_{j=3}^{n}\frac{d^3\mathbf{p}_j}{(2\pi)^3 2p_0^j}\cdot\frac{|m_{\alpha\beta}|^2(2\pi)^4}{4\sqrt{(P_1\cdot P_2)^2 - m_1^2 m_2^2}}\,\delta^4(P_3 + P_4 + \cdots - P_1 - P_2)}$

$\underbrace{(2\pi)^4\delta^4(P_\alpha - P_\beta)}_{V\cdot T}\,(2\pi)^4\delta^4(P_\alpha - P_\beta)\,|m_{\alpha\beta}|^2$

①

問2 $1+2 \to 3+4$ (2体散乱) に①が $d\sigma = \dfrac{1}{64\pi^2}\cdot\dfrac{|\mathbf{p}'|}{(P_1^0 + P_2^0)^2|\mathbf{p}|}|m_{\alpha\beta}|^2 d\Omega$ を与えることを示せ

重心系で

$\mathbf{p}_1 = -\mathbf{p}_2 = \mathbf{p}$

$(\mathbf{p}_3 = -\mathbf{p}_4 = \mathbf{p}')$

$\mathbf{p}'$ の $\mathbf{p}$ に対する3次元立体角 $d\Omega$

問1. $\exists a\mathbf{v}_1 + \exists b\mathbf{v}_2 = 0$ なるLorentz系で

$\bar{v}_{rel} = |\mathbf{v}_1 - \mathbf{v}_2|$ を示せ

COLLEGE NOTES / n056

崩壊確率

$M \to \overbrace{1+2+\cdots+n}^{\beta}$

$|\alpha\rangle$ --- 1粒子 $P_\alpha^\mu = (M, 0)$ $\Rightarrow M$

不変規格化 $\to$ 全空間に $2P^0V = 2MV$ 個

$$dN = \prod_{j=1}^{n} \frac{d^3P_j}{(2\pi)^3 2P_{0j}} |\langle\beta|S|\alpha\rangle|^2$$

$$\langle\beta|S|\alpha\rangle = i(2\pi)^4\delta^4(P_\beta - P_\alpha)\, m_{\beta\alpha}$$

$$(2\pi)^4\delta^4(P_\beta - P_\alpha)\cdot \underbrace{(2\pi)^4\delta^4(0)}_{VT} |m_{\beta\alpha}|^2 \;/\; 2MV\cdot T$$

単位時間当り，親粒子が1個当り

$$d\Gamma = \frac{1}{2M}\prod_{j=1}^{n}\frac{d^3P_j}{(2\pi)^3 2P_{0j}}(2\pi)^4\delta^4(P_1+\cdots+P_n - M)|m_{\alpha\beta}|^2$$

特に2粒子崩壊の場合

$P_1 = -P_2 = P$

$$\boxed{d\Gamma = \frac{|P|}{8\pi M^2}|m_{\alpha\beta}|^2 \frac{d\Omega}{4\pi}}$$

$S = T\exp i\int d^4x\, \mathcal{L}_{int}$

$\mathcal{L}_{int} = e\bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)$

4. QED 散乱

$\frac{e^2}{4\pi} = \frac{1}{137}$

4.1 Møller 散乱

$e + e \to e + e$

$P_1\ P_2\ P_3\ P_4$

$s_1\ s_2\ s_3\ s_4$

$$\psi(x) = \int\frac{d^3P}{(2\pi)^3 2P_0}\sum_s\left(b(P,s)u(P,s)e^{-ipx} + d^\dagger(P,s)v(P,s)e^{ipx}\right)$$

$$A_\mu(x) = \cdots\, a(k,\sigma)\varepsilon^{(\sigma)}(k)e^{-ikx} + a^\dagger(k,\sigma)\varepsilon^{(\sigma)*}(k)e^{ikx})$$

$\langle(P_4,s_4)(P_3,s_3)|S|(P_1,s_1)(P_2,s_2)\rangle$

$= \langle 0|B(P_4,s_4)B(P_3,s_3)\underset{\shortparallel}{S}B^\dagger(P_1,s_1)B^\dagger(P_2,s_2)|0\rangle$

$\langle P|q\rangle = 2P^0(2\pi)^3\delta^3(P-q)$

$B(P,s) = \sqrt{(2\pi)^3 2P_0}\, b(P,s)$

$T\exp i\left(\int e\bar{\psi}\gamma^\mu\psi A_\mu\right)$

$\frac{1}{2!}\left(i\int e\bar{\psi}\gamma^\mu\psi A_\mu\right)\left(i\int e\bar{\psi}\gamma^\nu\psi A_\nu\right)$

$$= \int i d^4x\int i d^4y\,\frac{1}{2!}\langle 0|B_4B_3\,T\,e\bar{\psi}(2)\gamma^\mu\psi(x)\,e\bar{\psi}(y)\gamma^\nu\psi(y)\,B_1^\dagger B_2^\dagger|0\rangle \times\langle 0|T A_\mu(x)A_\nu(y)|0\rangle$$

① ② $b^\dagger+d$ $b+d^\dagger$ $b^\dagger+d$ $b+d^\dagger$

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$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$

$\not{a}\not{b} + \not{b}\not{a} = 2(a\cdot b)$

$\{ b(p_1,s_1), b^\dagger(p_2,s_2) \} = \delta^3(p_1-p_2)\,\delta_{s_1 s_2}$

$\langle 0|\psi(x)\,B_2^\dagger|0\rangle = u(p_2,s_2)\,e^{-ip_2x}$ $\qquad$ $\langle 0|B_4\,\bar\psi(x)|0\rangle = \bar u(p_4,s_4)\,e^{ip_4x}$

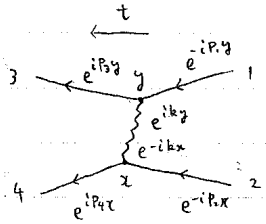
$= \int d^4x \int d^4y \Big[+\bar u_4 e^{ip_4x} e\gamma^\mu u_2 e^{-ip_2x} \cdot \bar u_3 e^{ip_3y} e\gamma^\nu u_1 e^{-ip_1y} \quad \cdots ①$

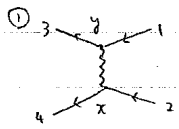
$\quad -\bar u_3 e^{ip_3x} e\gamma^\mu u_2 e^{-ip_2x} \cdot \bar u_4 e^{ip_4y} e\gamma^\nu u_1 e^{-ip_1y} \Big] \quad \cdots ②$

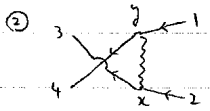
$\times \int \frac{d^4k}{i(2\pi)^4} \cdot \frac{g_{\mu\nu}}{k^2}\, e^{-ik(x-y)}$

$e\bar\psi\gamma^\mu\psi A$









$\langle\beta|S|\alpha\rangle = 1 + i(2\pi)^4\,\delta^4(p_\beta - p_\alpha)\,\mathcal{M}_{\beta\alpha}$

$e^{i(p_3-p_1+k)y} = i(2\pi)^4\,\delta^4(p_3-p_1+k)$

$i(2\pi)^4\,\delta^4(p_4-p_2-k)$ $\qquad$ $\mathcal{M} = \Big(\bar u_4 e\gamma^\mu u_2 \cdot \bar u_3 e\gamma^\nu u_1 \frac{g_{\mu\nu}}{(p_1-p_3)^2} - \bar u_3 e\gamma^\mu u_2 \cdot \bar u_4 e\gamma^\nu u_1 \frac{g_{\mu\nu}}{(p_1-p_4)^2} \Big)$

$d\sigma \propto |\mathcal{M}|^2$ $\qquad$ $= e^2 \Big(\bar u_4 \gamma^\mu u_2 \cdot \bar u_3 \gamma_\mu u_1 \frac{1}{(p_1-p_3)^2} - \bar u_3 \gamma^\mu u_2 \cdot \bar u_4 \gamma_\mu u_1 \frac{1}{(p_1-p_4)^2} \Big)$

① $(\bar u_3 \Gamma u_1)^* = \bar u_1 \bar\Gamma u_3$ $\qquad$ $\bar\Gamma \equiv \gamma_0\Gamma^\dagger\gamma_0$ $\qquad$ $\bar\gamma_\mu = \gamma_\mu$ $\qquad$ ${}^t x A x = \mathrm{tr}[A(\underset{n\times n}{x}\,{}^t x)]$ $\quad {}_{n\times m}$

② $(\bar u_4 \gamma^\mu u_2 \cdot \bar u_3 \gamma_\mu u_1) \cdot (\bar u_2 \gamma^\nu u_4 \cdot \bar u_1 \gamma_\nu u_3)$

$= \bar u_4 \gamma^\mu (u_2 \bar u_2) \gamma^\nu u_4 \cdot \bar u_3 \gamma_\mu u_1 \cdot \bar u_1 \gamma_\nu u_3$

4×4 $\quad (\cdots)^{(\cdots)}$

$= \mathrm{tr}\left[\gamma^\mu (u_2\bar u_2)\gamma^\nu (u_4\bar u_4)\right] \cdot \mathrm{tr}\left[\gamma_\mu (u_1\bar u_1)\gamma_\nu (u_3\bar u_3)\right]$

$\boxed{\frac{1}{4}\sum_{s_1 s_2}\sum_{s_3 s_4}}$ ← initial state spin 平均 / final state spin 足し上げ

▷ Jul 15

$\overline{|(\bar u_3 \gamma^\mu u_1)(\bar u_4 \gamma_\mu u_2)|^2} = \bar u_3 \gamma^\mu u_1 \bar u_4 \gamma_\mu u_2 \bar u_1 \gamma^\mu u_3 \bar u_2 \gamma_\mu u_4$

$= \mathrm{tr}(\gamma^\mu u_1 \bar u_1 \gamma^\nu u_3 \bar u_3) \times \mathrm{tr}(\gamma_\mu u_2 \bar u_2 \gamma_\nu u_4 \bar u_4)$

$\sum_s u(p,s)\bar u(p,s) = \not p + m$

$= \mathrm{tr}\left[\gamma^\mu(\not p_1 + m)\gamma^\nu(\not p_3 + m)\right] \times \mathrm{tr}\left[\gamma_\mu(\not p_2 + m)\gamma_\nu(\not p_4 + m)\right]$

$= \mathrm{tr}\left[\gamma^\mu \not p_1 \gamma^\nu \not p_3 + m^2\gamma^\mu\gamma^\nu\right] \cdot \mathrm{tr}\left[\gamma_\mu \not p_2 \gamma_\nu \not p_4 + m^2\gamma_\mu\gamma_\nu\right]$

$= 4\left(p_1^\mu p_3^\nu - g^{\mu\nu}(p_1\cdot p_3) + p_3^\mu p_1^\nu + m^2 g_{\mu\nu}\right) \times$

$4\left(p_{2\mu}p_{4\nu} - g_{\mu\nu}\,p_2\cdot p_4 + p_{4\mu}p_{2\nu} + m^2 g_{\mu\nu}\right)$

$= 32\left[(p_1\cdot p_3)^2 + (p_1\cdot p_4)^2 + 2m^2(m^2 - (p_1\cdot p_3))\right]$

$p_1+p_2 = p_3+p_4 \rightarrow p_1\cdot p_2 = p_3\cdot p_4$

$p_1-p_4 = p_3-p_2 \rightarrow p_1\cdot p_4 = p_3\cdot p_2$

公式

$\mathrm{tr}\,1 = 4$

$\mathrm{tr}(\not a\not b) = 4(a\cdot b)$

$\mathrm{tr}(\not a\not b\not c\not d) = 4\left[(a\cdot b)(c\cdot d) - (a\cdot c)(b\cdot d) + (a\cdot d)(b\cdot c)\right]$

tr(奇数個) $= 0$

Cross term

$= -(\bar{u}_3 \gamma^\mu u_1 \bar{u}_4 \gamma_\mu u_2 \cdot \bar{u}_1 \gamma^\nu u_4 \bar{u}_2 \gamma_\nu u_3 + c.c)$

$= -\mathrm{tr}\left(\gamma^\mu (u_1\bar{u}_1)\gamma^\nu (u_4\bar{u}_4)\gamma_\mu (u_2\bar{u}_2)\gamma_\nu (u_3\bar{u}_3)\right) - c.c.$

$= -\mathrm{tr}\left\{\gamma^\mu (\not{p}_1+m)\gamma^\nu (\not{p}_4+m)\gamma_\mu (\not{p}_2+m)\gamma_\nu (\not{p}_3+m)\right\} - c.c$

公式

$\gamma^\mu \not{a} \gamma_\mu = -2\not{a}$

$\gamma^\mu \not{a} \not{b} \gamma_\mu = 4\,a\cdot b$

$\gamma^\mu \not{a} \not{b} \not{c} \gamma_\mu = -2\not{c}\not{b}\not{a}$

$\gamma^\mu \left[\not{p}_1\gamma^\nu \not{p}_4 + m(\gamma^\nu \not{p}_4 + \not{p}_1\gamma^\nu) + m^2\gamma^\nu\right]\gamma_\mu$

$(\underbrace{-2\not{p}_4\gamma^\nu \not{p}_1}_{①} + \underbrace{4m(p_4^\nu + p_1^\nu)}_{②} - \underbrace{2m^2\gamma^\nu}_{②})(\not{p}_2+m)\gamma_\nu(\not{p}_3+m)$

① $2\,\mathrm{tr}\left[\not{p}_4\gamma^\nu \not{p}_1(\not{p}_2+m)\gamma_\nu(\not{p}_3+m)\right]$

$\gamma^\nu(\not{p}_1\not{p}_2 + m\not{p}_1)\gamma_\nu$

$= 2\,\mathrm{tr}\left[\not{p}_4(4p_1\cdot p_2 - 2m\not{p}_1)(\not{p}_3+m)\right]$

$= 2\cdot 4\left[(p_3\cdot p_4)4(p_1\cdot p_2) - 2m^2(p_1\cdot p_4)\right]$

② $-4m(p_1^\nu + p_4^\nu)\,\mathrm{tr}\left[\underbrace{(\not{p}_2+m)\gamma_\nu(\not{p}_3+m)}\right] \quad 4m(p_{3\nu}+p_{3\nu})$

$= -16m^2(p_1+p_4)\cdot(p_2+p_3)$

③ $2m^2\,\mathrm{tr}\left(\underbrace{\gamma^\nu(\not{p}_2+m)\gamma_\nu}_{(-2\not{p}_2+4m)}(\not{p}_3+m)\right)$

$= 2m^2\cdot 4(-2p_2\cdot p_3 + 4m^2)$

$p_1\cdot p_2 = p_3\cdot p_4$

$p_1\cdot p_3 = p_2\cdot p_4$

$p_2\cdot p_3 = p_1\cdot p_4$

$= 8\left[4(p_1\cdot p_2)^2 - 2m^2 p_1\cdot p_4 - 4m^2(p_1\cdot p_2 + p_1\cdot p_3) - 2m^2(\overbrace{p_2\cdot p_3}^{p_1\cdot p_4}) + 4m^2\right]$

$= 8\left[4(p_1\cdot p_2)^2 - 4m^2 p_1\cdot(\underbrace{p_4+p_2+p_3}_{p_1+2p_2}) + 4m^2\right]$

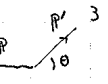
$= 32\left[(p_1\cdot p_2)^2 - 2m^2(p_1\cdot p_2)\right]$

$\downarrow \times 2$ (① c.c)

$$\frac{1}{4}\sum_{\substack{s_1 s_2\\ s_3 s_4}} |m|^2 = \frac{32}{4}e^4\left[\frac{(p_1\cdot p_2)^2 + (p_1\cdot p_4)^2 - 2m^2(p_1\cdot p_3 - m^2)}{(p_1-p_3)^4} + \frac{(p_1\cdot p_2)^2 + (p_1\cdot p_3)^2 - 2m^2(p_1\cdot p_4 - m^2)}{(p_1-p_4)^4}\right.$$

$$\left. + 2\cdot\frac{(p_1\cdot p_2)^2 - 2m^2(p_1\cdot p_2)}{(p_1-p_3)^2(p_1-p_4)^2}\right]$$

(全て) in CM



$p_1 = (E, \mathbf{p}) \quad E = \sqrt{\mathbf{p}^2+m^2}$

$p_2 = (E, -\mathbf{p})$

$p_3 = (E, \mathbf{p}')$

$p_4 = (E, -\mathbf{p}')$

$p_1\cdot p_2 = \frac{1}{2}\left((p_1+p_2)^2 - 2m^2\right) = 2E^2 - m^2$

$p_1\cdot p_3 = E^2(1-\cos\theta) + m^2\cos\theta$

$p_1\cdot p_4 = E^2(1+\cos\theta) - m^2\cos\theta$

$\mathbf{p}\cdot\mathbf{p}' = \mathbf{p}^2\cos\theta = (E^2-m^2)\cos\theta$

Møller 公式 $\alpha \equiv \frac{e^2}{4\pi}$

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2(2E^2-m^2)^2}{4E^2(E^2-m^2)^2}\left[\frac{4}{\sin^4\theta} - \frac{3}{\sin^2\theta} + \frac{(E^2-m^2)^2}{(2E^2-m^2)^2}\left(1+\frac{4}{\sin^2\theta}\right)\right]$$

超相對論的極限

$m/E \ll 1$ ⟶ $\frac{\alpha^2}{E^2}\left(\frac{4}{\sin^4\theta} - \frac{2}{\sin^2\theta} + \frac{1}{4}\right) = \frac{\alpha^2}{4E^2}\left(\frac{1}{\sin^4\frac{\theta}{2}} + \frac{1}{\cos^4\frac{\theta}{2}} + 1\right)$

非相對論的極限

$E \simeq m + v^2/2m$ ⟶ $\left(\frac{\alpha}{m}\right)^2 \cdot \frac{1}{4v^4}\left(\frac{4}{\sin^4\theta} - \frac{3}{\sin^2\theta}\right) = \left(\frac{\alpha}{m}\right)^2 \cdot \frac{1}{16v^4}\left(\frac{1}{\sin\frac{\theta}{2}} + \frac{1}{\cos^4\frac{\theta}{2}} - \frac{1}{\sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2}}\right)$

$v^2 \simeq \frac{E^2-m^2}{E^2}$

問 1. $e + \mu \to e + \mu$ (p_1 p_2; m_e / m_μ) $m_\mu \cong 100\ MeV$, $m_e \cong 0.5\ MeV$

= cross section in Lab 系 (initial μ at rest frame)

$\begin{cases} P_2 = (m_\mu, 0) \\ P_1 = (E, 0, 0, \sqrt{E^2-m_e^2}) \end{cases}$

2" 計算

Bhabha 散亂 $e + \bar{e} \to e + \bar{e}$

p_1 q_2 p_3 q_4

s_1 s_2 s_3 s_4

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5. Weak decay $\pi^{\pm} \to \ell^{\pm} + \nu$

QED $\mathcal{L}_{int} = e\bar{\Psi}\gamma^{\mu}\Psi A_{\mu}$

4-fermi interaction $\Psi_e, \Psi_\mu, \Psi_{\nu_e} \to e, \mu, \nu_e$ 粒子

$$\mathcal{L}_{Fermi} = -\frac{G}{\sqrt{2}} J_{\mu}^{\dagger} J^{\mu}$$

1967 Weinberg-Salam

charge current $J_\mu = J_\mu^{(lepton)} + J_\mu^{(quark)}$

$$J_\mu^{(lepton)} = \bar{e}\gamma_\mu(1-\gamma_5)\nu_e + \bar{\mu}\gamma_\mu(1-\gamma_5)\nu_\mu + \bar{\tau}\gamma_\mu(1-\gamma_5)\nu_\tau$$

$$J_\mu^{(quark)} = \bar{d}'\gamma_\mu(1-\gamma_5)u + \bar{s}'\gamma_\mu(1-\gamma_5)c + \bar{b}'\gamma_\mu(1-\gamma_5)t$$

$Q = +\frac{2}{3}$ u c (t)

$Q = -\frac{1}{3}$ d s b $\quad SO(2)$

Cabibbo mixing

$$\begin{pmatrix} d' \\ s' \end{pmatrix} = \begin{pmatrix} \cos\theta_c & \sin\theta_c \\ -\sin\theta_c & \cos\theta_c \end{pmatrix}\begin{pmatrix} d \\ s \end{pmatrix}$$

$$\begin{pmatrix} d' \\ s' \\ b' \end{pmatrix} = \begin{pmatrix} \cup \\ \| \\ \text{CKM行列} \end{pmatrix}\begin{pmatrix} d \\ s \\ b \end{pmatrix}$$

$\pi^- \to \ell^{e,\mu} + \bar{\nu}_\ell$

$k \quad P_1 \quad P_2$

S_1

$|\pi^-(k)\rangle \sim D_\mu^\dagger B_d^\dagger |0\rangle$ ← 2粒子からなる π^- 中間 (bound state)

$B_\ell^\dagger(P_1, S_1) D_{\nu_e}^\dagger(P_2)|0\rangle$

$$i(2\pi)^4\delta^4(P_1+P_2-k)\,\mathcal{M} = \langle 0|D_{\nu_\ell}(P_2)B_\ell(P_1,S_1)\ \underset{\|}{S}\ |\pi^-(k)\rangle$$

$$\mathrm{T}\exp i\int d^4x\,\mathcal{L}_{int}$$

$$i\int d^4x\,-\frac{G}{\sqrt{2}}J_\mu^{\dagger(quark)}J^{\mu(lepton)}$$

$i\dot{\theta}(t) = [P_0, \theta(t)]$

$-i\partial_\mu\theta(x) = [P_\mu, \theta(x)]$

$\theta(x) = e^{iP\cdot x}\theta(0)e^{-iP\cdot x}$

$= \theta(0) + i[P_x, \theta]$

$$= i\int d^4x\left(-\frac{G}{\sqrt{2}}\right)\langle 0|D_{\nu_e}(P_2)B_\ell(P_1,S_1)\,\bar{\ell}(x)\gamma_\mu(1-\gamma_5)\nu_\ell(x)|0\rangle \times \cos\theta_c\langle 0|\bar{u}\gamma_\mu(1-\gamma_5)d(x)|\pi^-(k)\rangle$$

$$= \left(-\frac{G}{\sqrt{2}}\cos\theta_c\right)\int i d^4x\,\bar{u}_\ell(P_1,S_1)e^{iP_1x}\gamma_\mu(1-\gamma_5)v_\nu(P_2)e^{iP_1x} \times \sqrt{2}f_\pi\, ik^\mu e^{-ikx}$$

$\langle 0|\bar{u}(x)\gamma_\mu(1-\gamma_5)d(x)|\pi^-(k)\rangle = \langle 0|e^{iPx}\bar{u}(0)\gamma_\mu(1-\gamma_5)d(0)e^{-iP\cdot x}|\pi^-(k)\rangle \qquad k_0 = \sqrt{k^2+m_\pi^2}$

$$= \underline{\langle 0|\bar{u}(0)\gamma_\mu(1-\gamma_5)d(0)|\pi^-(k)\rangle}\,e^{-ikx}$$

$\|$

$ik_\mu\boxed{\sqrt{2}f_\pi}$ $\quad f_\pi$：πの崩壊定数

$$= i(2\pi)^4\delta^4(P_1+P_2-k)\times\cdots \qquad k = k^\mu\gamma_\mu$$

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$\mathcal{M}_\ell = -i G_\pi \bar{u}_\ell \not{k}(1-\gamma_5) v_\nu = -i G_\pi m_\ell \bar{u}_\ell (1-\gamma_5) v_\nu$

$G_\pi \equiv G f_\pi \cos\theta_c$ 　　$k = p_1 + p_2$ (p_1: m_ℓ, p_2: 0)

$k = p_1 + p_2$ 　　$(\not{p}_1 - m_\ell) u_\ell(p_1, s_1) = 0$

$\bar{u}_\ell (\not{p}_1 - m_\ell) = 0$

$(\not{p}_2 + 0) v_\nu(p_2) = 0$

decay width

$$\Gamma(\pi^- \to \ell + \bar{\nu}_\ell) = \frac{|\vec{p}|}{8\pi m_\pi^2} \int \underbrace{\sum_{s_i} |\mathcal{M}_\ell|^2}_{\text{この計算をせよ」… 問1(1)}} \frac{d\Omega}{4\pi}$$

$$= \frac{(G f_\pi \cos\theta_c)^2}{4\pi m_\pi^3} m_\ell^2 (m_\pi^2 - m_\ell^2)^2$$

$$\frac{\Gamma(\pi^- \to e + \bar{\nu}_e)}{\Gamma(\pi^- \to \mu + \bar{\nu}_\mu)} = \frac{m_e^2 (m_\pi^2 - m_e^2)}{m_\mu^2 (m_\pi^2 - m_\mu^2)} = \underline{1.28 \times 10^{-4}}$$

$m_e = 0.5$ MeV 　$m_\mu = 106$ MeV 　$m_\pi = 140$ MeV

$f_\pi = 93$ MeV 　$G = 1.17 \times 10^{-5}$ (GeV)$^{-2}$ 　$\theta_c = 0.23$

$\pi^- \to \mu + \bar{\nu}_\mu$ 　$\sim 100\%$ 　100%とみなして π^- の寿命 τ を数値 (sec) で答えよ」… 問1(2)

$\bar{J}_\mu^{(\text{electron})} = \bar{e} \gamma_\mu (1-\gamma_5) \nu = 2 \bar{e}_L \gamma_\mu \nu_L$ 　　$\nu_L = \frac{1-\gamma_5}{2}\nu$

$= 2(\bar{\nu}_L, \bar{e}_L) \gamma_\mu \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$ 　　$e_L = \frac{1-\gamma_5}{2} e$

$\tau^- = \frac{\tau^1 - i\tau^2}{2}$ 　　P_L 　$P_L^2 = P_L$

$L \equiv \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$

$= 2 \bar{L} \gamma_\mu \frac{\tau^1 - i\tau^2}{2} L = 2(j_\mu^1 - i j_\mu^2)$ 　τ は (1°う) 行列

$j_\mu^a \equiv \bar{L} \gamma_\mu \frac{\tau^a}{2} L$ 　SU(2) current 　$a = 1.2.3$

$Q^a = \int d^3x \, j_{\mu=0}^a(x)$

$[Q^a, Q^b] = i \epsilon_{abc} Q^c$ 　SU(2) 代数

isospin

if 理論 local SU(2) 不変 ; 　$L(x) \to L'(x) = \underbrace{e^{i\theta^a(x)\frac{\tau^a}{2}}}_{SU(2)} L(x)$

$\mathcal{L} = \bar{L} i\gamma^\mu (\partial_\mu - i g \underbrace{A_\mu^a(x)}_{\text{〰}} \frac{\tau^a}{2}) L - \frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu}$ 　$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g \epsilon_{abc} A_\mu^b A_\nu^c$

$\mathcal{L}_{int} = g A_\mu^a \bar{L} \gamma^\mu \frac{\tau^a}{2} L$

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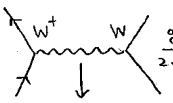
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$$\mathcal{L}_{int} = g A_\mu^a \bar{L} \gamma^\mu \frac{\tau^a}{2} L \qquad (\bar{\nu}_L \ \bar{e}_L)\gamma^\mu \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}$$

$$= \frac{1}{2} g \left[\underbrace{(A_\mu^1 + i A_\mu^2)}_{\sqrt{2}\, W_\mu^+} \cdot \underbrace{\bar{L} \gamma^\mu \frac{\tau^1 - i\tau^2}{2} L}_{\frac{1}{2} J^\mu} + h.c. \right] + \underbrace{g A_\mu^3}_{\text{photon}} \cdot \bar{L} \gamma^\mu \frac{\tau^3}{2} L \qquad \Leftarrow -\frac{1}{2} \bar{e}_R \gamma_\mu e_R$$

$$\left(\frac{1}{2} \bar{\nu}_L \gamma_\mu \nu_L - \frac{1}{2} \bar{e}_L \gamma_\mu e_L \right)$$

$$= \frac{g}{2\sqrt{2}} \left(W_\mu^+ J^\mu + W_\mu J^{\mu +} \right) + g A_\mu^3 \cdot \bar{L} \gamma^\mu \frac{\tau^3}{2} L$$



$$\frac{g}{2\sqrt{2}} J_\mu \qquad \frac{g}{2\sqrt{2}} J_\nu^+ \simeq -\frac{g^2}{8 M_W^2} J_\mu J^{\mu +}$$

$p^2 \ll M_W^2$ $\qquad \frac{g^{\mu\nu}}{p^2 - M_W^2}$ $\qquad \frac{G}{\sqrt{2}}$

$$M_W^2 = \frac{g^2 \sqrt{2}}{8G} \overset{10^{-2} \sim \left(\frac{g^2}{4\pi}\right)\left(\sqrt{2}\, 4\pi\right) \sim 20}{=} \frac{}{8G} = \left(\frac{1}{4} \times 10^4\right) (GeV)^2$$

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